# Can an MLP Absorb Its Skip Connection Exactly?

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**Abstract**

Residual networks add each block's input to its output; skipless ones now train as well. Can a block without biases absorb its skip: can a skipless block of the same kind compute the same function? Here we show that almost every ReLU or GELU block of at least $d_{\text{model}}$ units absorbs its skip with exactly $d_{\text{model}}$ more units, with $d_{\text{model}}$ the model dimension, while no number of squared-ReLU, gated (ReGLU, SwiGLU, GeGLU) or $\tanh$ units or attention heads does. Without normalization, no skipless squared-ReLU, gated or $\tanh$ network equals a residual one, and a skipless ReLU or GELU network needs at least $2d_{\text{model}}$ units in its first block to equal almost any residual one; yet every skipless network is a limit of residual ones of equal widths. Under RMSNorm, almost no residual network equals any skipless one. These results say when a skip can be traded for hidden units.

## 1 Introduction

Almost every deep network trained today, from residual networks (He et al., 2016) to transformers (Vaswani et al., 2017), adds the input of each block to its output through a skip connection. The usual reason is training: the sum gives the gradient a short path through the network, and it makes the identity map easy to express (Hardt and Ma, 2017; Veit et al., 2016).

Skipless networks now train too. He and Hofmann (2024) remove the skips of transformer blocks without losing training speed, and Ji et al. (2026) train a skipless ViT-B on ImageNet that outperforms the residual one. Can a skip let a block compute a function that no skipless block of the same kind computes?

We answer this question block by block. Take an MLP block $B$ with $m$ hidden units, or an attention block with $H$ heads, on hidden states of dimension $d_{\text{model}}$; its width is $m$ or $H$. Two blocks are of the same kind if both are vanilla MLP blocks with the same activation, both are gated with the same gate, or both are attention blocks. Is the residual block $x \mapsto x + B(x)$ also a block of the same kind without a skip, of the same width or wider? If it is, we say that $B$ *absorbs* its skip. The skip is worth $n$ units if $B$ absorbs it with $n$ extra units but not with $n-1$; if no width suffices, it is worth more than any number of units.

**Without biases, the answer depends on the activation.**

- For almost every ReLU or GELU block with at least $d_{\text{model}}$ units, a skip is worth exactly $d_{\text{model}}$ units: $d_{\text{model}}$ new units, each mirroring an existing unit, absorb it, and fewer do not (Theorem 3.1(ii)).
- For ReLU$^2$ blocks, whose activation is the square of ReLU, gated blocks (ReGLU, SwiGLU, GeGLU) and $\tanh$ blocks, no number of units absorbs a skip (Theorem 3.3).
- For attention blocks, no number of heads absorbs a skip (Theorem 3.4).

*Author contributions.* Marko Karbevski (MK) conceived and led the study. Antonij Mijoski (AM) proved the absorption result for a single ReLU block (Theorem 3.1(i), Theorem A.5), the difficult case, paving the way forward. MK extended the tools to GELU (Theorem A.7), proved the cases of ReLU$^2$ and of the gated variants ReGLU, SwiGLU and GeGLU (Theorems 3.3 and 4.1) and the results at depth (Theorem 4.2 and Proposition 4.3). Both authors worked on the writing. All of MK's work on the results and the proofs was done with AI assistants.

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Outside Appendix C, blocks have no biases, as in many current language models (Team OLMo et al., 2025; Yang et al., 2025). With biases, every ReLU$^2$ and gated block absorbs its skip with more units, while $\tanh$ and attention blocks still never do (Appendix C).

**Four one-line reasons.** Follow the input along a ray, $x = tu$ with $t > 0$. The skip adds $tu$, which grows like $t$. A ReLU$^2$ block grows like $t^2$. A gated block, whose gate vanishes at 0, is of order $t^2$ near the origin. A $\tanh$ block stays bounded. So none of them can produce the term $tu$ (Figure 1, Section 3.2). Attention mixes the tokens of a sequence, while a skip leaves each token where it is.

**Networks.** Throughout, a network starts with a linear embedding and ends with a linear unembedding. In a pre-RMSNorm network, every block and the unembedding read their input through RMSNorm, which divides each token by its root mean square, regularized by a constant $\varepsilon \geq 0$. Without normalization, no residual network of ReLU$^2$, gated or $\tanh$ blocks equals a skipless network of any widths (Theorem 4.1), and a skipless ReLU or GELU network needs at least $2d_{\text{model}}$ units in its first block, whatever its later widths, to equal almost any residual one (Theorem 4.2). Under RMSNorm, almost no residual network, of any blocks, equals a skipless network of any widths (Theorem 4.4): as an input is scaled by $t \to \infty$, the output of a skipless network converges at rate $1/t^2$ or faster, and that of almost every residual network only at rate $1/t$.

**No function lost in the limit.** Without normalization, every skipless network is a limit of residual networks of the same widths, uniformly on bounded sets of inputs, as their weights diverge (Proposition 5.1(i)): the skip adds functions and, in the limit, removes none. Table 1 collects the results; Section 2 fixes the setting, Section 3 treats one block, Section 4 networks and Section 5 limits. Proofs are in the appendix.

## 2 Setup

*One equation states the question: can a block absorb a linear term?*

**Blocks.** A hidden state is a matrix $h \in \mathbb{R}^{T \times d_{\text{model}}}$ whose rows are the $T$ tokens of a sequence. An MLP block acts on each token separately. On a row $x \in \mathbb{R}^{1 \times d_{\text{model}}}$ of a hidden state it computes

$$B_\theta(x) = f(xW_{\text{read}})\, W_{\text{write}}, \qquad \theta = (W_{\text{read}}, W_{\text{write}}), \tag{1}$$

with a read $W_{\text{read}}$ and a write $W_{\text{write}}$. In a vanilla block with $m$ hidden units, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$, $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, and $f$ applies an activation $\sigma$ to each unit: ReLU, GELU, $\tanh$, or the square of ReLU, ReLU$^2$. In a gated block the read has two halves, $W_{\text{read}} = [\, W_{\text{g}} \mid W_{\text{v}} \,]$, and unit $i$ multiplies its gate $\sigma\big((xW_{\text{g}})_i\big)$ by its value $(xW_{\text{v}})_i$, with $\sigma$ the ReLU in ReGLU, SiLU in SwiGLU and GELU in GeGLU (Shazeer, 2020). Attention blocks follow in Section 3.3.

**The question as an equation.** A skip adds $x$ to the output of a block. More generally, can a block absorb a linear term $xM$, for a $d_{\text{model}} \times d_{\text{model}}$ matrix $M \neq 0$? We look for parameters $\theta'$ of a block of the same kind, and of any width, such that

$$xM + B_\theta(x) = B_{\theta'}(x) \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{2}$$

The skip is $M = I$. Other matrices cover learned linear skips and, below, networks.

**Networks.** A network maps an input $X \in \mathbb{R}^{T \times d_{\text{in}}}$ to

$$\mathcal{M}(X) = C \circ D \circ (A_L + B_L) \circ \cdots \circ (A_1 + B_1) \circ E\,(X),$$

where $E$ is the linear embedding $X \mapsto XE$, with $E \in \mathbb{R}^{d_{\text{in}} \times d_{\text{model}}}$, $B_1, \ldots, B_L$ are $L \geq 1$ blocks, $D$ is the linear readout, or unembedding, $h \mapsto hD$, with $D \in \mathbb{R}^{d_{\text{model}} \times d_{\text{out}}}$, $A_\ell$ is the skip of block $\ell$, the zero map in a *skipless* network and the identity map $I$ in a *residual* one, and $C$ is an output function set by the task: the identity for regression, with one output or several, the logistic sigmoid of each output for multi-label classification, and the softmax for multiclass classification (Bishop, 2006, Sections 5.1 and 5.2). So the hidden states are $h_0 = XE$ and $h_\ell = B_\ell(h_{\ell-1})$ in a skipless network and $h_\ell = h_{\ell-1} + B_\ell(h_{\ell-1})$ in a residual one, the readout gives $y = h_L D$, and

| **Proven** | **ReLU** | **GELU** | **tanh** | **ReLU$^2$, ReGLU** | **SwiGLU, GeGLU** | **Attention** |
|---|---|---|---|---|---|---|
| *Can a skipless network equal a residual one? Without normalization:* | | | | | | |
| one block, same width | measure zero Thm. 3.1(i) | | never Thm. 3.3 | | | never Thm. 3.4 |
| one block, more hidden units or heads | exactly $d_{\text{model}}$ more Thm. 3.1(ii) | | no Thm. 3.3 | | | no Thm. 3.4 |
| networks of one block, same width | measure zero Prop. 3.2 | | never Thm. 4.1(ii) | never Thm. 4.1(i) | | never Thm. A.12(iii) |
| deeper networks, same depth and widths | measure zero if the first block has fewer than $2d_{\text{model}}$ units, else unknown Thm. 4.2 | measure zero companion | never Thm. 4.1(ii) | never Thm. 4.1(i) | | measure zero companion |
| networks, more hidden units or heads per block | generically, $d_{\text{model}}$ more suffice Prop. 4.3, and the first block needs $2d_{\text{model}}$ Thm. 4.2 | | no Thm. 4.1(ii) | no Thm. 4.1(i) | | Unknown |
| *With pre-RMSNorm:* | | | | | | |
| networks, $\varepsilon > 0$ | measure zero, at any width Thm. 4.4 | | | never Thm. 4.1(i) | | measure zero, at any width Thm. 4.4 |
| networks, $\varepsilon = 0$ | measure zero, at any width Thm. 4.4 | | | | | |
| *Is every skipless network a limit of residual ones of the same widths?* | | | | | | |
| without normalization | yes Prop. 5.1(i) | | | | | |
| with pre-RMSNorm, $\varepsilon = 0$ | yes Prop. 5.1(ii) | | | | | |
| with pre-RMSNorm, $\varepsilon > 0$ | Unknown | | | | | |

**Table 1:** What is proven, by setting. Never or no: for no weights. Measure zero: only for a null set of weights. Exactly $d_{\text{model}}$ more: for almost every block, $d_{\text{model}}$ more hidden units suffice and fewer do not. Companion: Karbevski and Mijoski (2026). No biases; scope in Section 2.

$\mathcal{M}(X) = C(y)$. The function $C$ has no weights, and we ignore it: equality of networks, and every result below, concern $y$. Networks with the same $y$ have the same $\mathcal{M}$, but not conversely; a softmax, for instance, does not change when the same number is added to every output. In a *pre-RMSNorm* network every block and the readout read the normalized state $N(h)$, while the skips carry $h$: in the display, $B_\ell$ and $D$ become $B_\ell \circ N$ and $D \circ N$, so the skipless and the residual network compute $h_\ell = B_\ell\big(N(h_{\ell-1})\big)$ and $h_\ell = h_{\ell-1} + B_\ell\big(N(h_{\ell-1})\big)$, and $y = N(h_L)D$. Here $N$ divides each row $v$ of $h$ by $\sqrt{\|v\|^2/d_{\text{model}} + \varepsilon}$, with $\varepsilon \geq 0$. For $\varepsilon = 0$, $N$ is defined on hidden states without zero rows, a network is defined at the inputs at which no state that it normalizes has a zero row, and two networks are equal if they agree on a dense set of inputs at which both are defined.

**No biases.** Blocks have no biases, and the answers depend on it. With biases, a gated unit whose gate read is zero has a constant gate, so it is affine in $x$, and $d_{\text{model}}$ such units absorb a skip. Appendix C shows that, with biases, every ReLU, GELU, ReLU$^2$ and gated block absorbs its skip, and no $\tanh$ or attention block does; almost every ReLU, GELU and gated block needs exactly $d_{\text{model}}$ more units. The gate matters even without biases: the logistic sigmoid gate of the original GLU does not vanish at 0, so a unit with a zero gate read is linear, and $d_{\text{model}}$ such units absorb a skip.

**Scope.** Every MLP block has at least $d_{\text{model}}$ hidden units, so that its read can have rank $d_{\text{model}}$. Residual networks have $ED \neq 0$. For simplicity, all MLP blocks of a network are of one kind, all attention blocks of a network have the same number of heads, and $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$. Most results hold more generally, and the appendix states them so. A statement holds for *generic* weights, or for almost every choice of weights, if it holds outside a set of Lebesgue measure zero.

# 3 One block

*For almost every ReLU or GELU block, a skip is worth exactly $d_{\text{model}}$ hidden units. For ReLU$^2$, gated,* $\tanh$ *and attention blocks it is worth more than any number.*

## 3.1 ReLU and GELU blocks: a skip is worth $d_{\text{model}}$ units

Both activations satisfy

$$\sigma(z) - \sigma(-z) = z, \tag{3}$$

so a unit and its mirror image, with the opposite read and the opposite write, add up to a linear term. This gives two ways to absorb $xM$.

*With more units.* Let unit $i$ have read $w_i$, a column of $W_{\text{read}}$, and write $\hat{w}_i$, a row of $W_{\text{write}}$. For a row $\gamma \in \mathbb{R}^{1\times d_{\text{model}}}$, a new unit with read $-w_i$ and write $-\gamma$, with the write of unit $i$ changed to $\hat{w}_i + \gamma$, adds $(xw_i)\,\gamma$ to the block, by (3). So each new unit adds one rank-one term, and $d_{\text{model}}$ new units, mirroring $d_{\text{model}}$ units whose reads span $\mathbb{R}^{d_{\text{model}}}$, add any $xM$.

*At the same width.* Flipping the read of unit $i$ to $-w_i$ changes the block by $-(xw_i)\,\hat{w}_i$. So $B_\theta$ absorbs $xM$ at the same width if some set $S$ of units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$, where $W_{\text{read}}[:, S]$ and $W_{\text{write}}[S, :]$ keep the columns and rows of the units in $S$.

The next theorem says that at the same width these flips are the only way, and that fewer than $d_{\text{model}}$ new units do not suffice. A block is *nondegenerate* if the columns of $W_{\text{read}}$ are nonzero and pairwise non-collinear and the rows of $W_{\text{write}}$ are nonzero.

**Theorem 3.1** (ReLU and GELU blocks)**.** *Let $\sigma$ be ReLU or GELU, and fix a matrix $M \neq 0$.*
**(i)** *If $B_\theta$ is nondegenerate, some block $B_{\theta'}$ of the same width satisfies (2) if and only if some set $S$ of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$. For generic $\theta$, no block of the same width does.*
**(ii)** *If $M$ is invertible and $W_{\text{read}}$ has rank $d_{\text{model}}$, some block $B_{\theta'}$ with $d_{\text{model}}$ more hidden units satisfies (2), and for generic $\theta$ none with fewer does.*

*Proof in Appendix A.1.*

In (i), for each of the finitely many sets $S$, the equation $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$ fails when the writes vanish, so it holds only on a set of weights of measure zero. By (ii), for almost every block the skip, $M = I$, is worth exactly $d_{\text{model}}$ units.

The proof matches the units of the two blocks. The even part of the activation, $\big(\sigma(z)+\sigma(-z)\big)/2$, matches each unit of the absorbing block to a unit of $B_\theta$, up to sign and scale: for ReLU through the kinks of $|z|$, and for GELU through the linear independence of ridge functions, which depend on $x$ only through one product $xw$. The odd part, $z/2$, turns each sign flip into a linear term, as Petzka et al. (2020, Section 3) show for two-layer ReLU networks with one output. Each extra unit then adds at most one rank-one term, and $M = I$ has rank $d_{\text{model}}$. Identity (3) alone is not enough. The activation $z/2 + |z|^3$ satisfies it, but scaling the read of a unit by $c > 0$ and its write by $c^{-3}$ keeps the even part of the unit and changes its linear part, which gives more ways to absorb $xM$ (Remark A.8).

**Networks of one block.** A residual network of one block is a block with the reads $EW_{\text{read}}$ and the writes $W_{\text{write}}D$, plus the linear term $XED$: it computes $X \mapsto XED + f(XEW_{\text{read}})\, W_{\text{write}}D$. The proof of Theorem 3.1(i) holds for such rectangular reads, writes and linear terms, which gives the following.

**Proposition 3.2** (Networks of one block)**.** *Let $\sigma$ be ReLU or GELU. For almost every choice of weights, a residual network of one block equals no skipless network of one block of the same width.*

*Proof in Appendix A.1.*

## 3.2 ReLU$^2$, gated and tanh blocks

Recall that a block absorbs $xM$ if a block of the same kind satisfies (2): $xM + B_\theta(x) = B_{\theta'}(x)$ for all $x$.

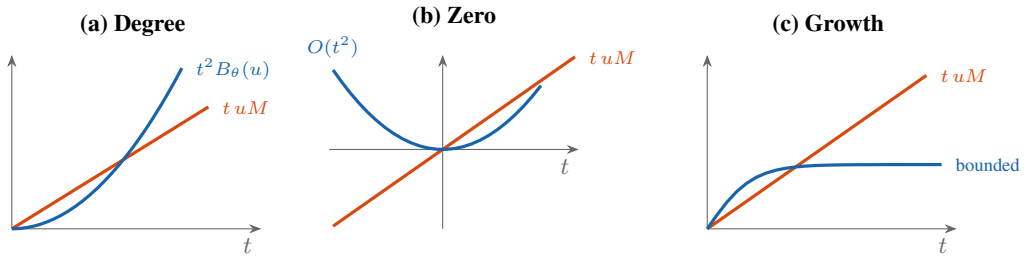



**Figure 1:** Why no width of ReLU$^2$, gated or $\tanh$ units absorbs the skip. Along a ray $x = tu$ the skip grows like $t$. (a) A ReLU$^2$ or ReGLU block grows like $t^2$. (b) A SwiGLU or GeGLU block is of order $t^2$ near $t = 0$. (c) A $\tanh$ block stays bounded.

**Theorem 3.3** (No width suffices)**.** *Let $B_\theta$ be a* $\tanh$*, ReLU$^2$, ReGLU, SwiGLU or GeGLU block, or more generally a block that is positively homogeneous of degree $k \neq 1$, or a gated block whose gate vanishes at* 0 *and is differentiable there. For every $M \neq 0$, no block $B_{\theta'}$ of the same kind and of any width satisfies* (2)*.*

*Proof in Appendix A.1.*

The three cases of Figure 1 are three one-line proofs. A homogeneous block satisfies $B_\theta(tu) = t^k B_\theta(u)$ for $t > 0$, so along a ray (2) reads $t\,uM = t^k\big(B_{\theta'}(u) - B_\theta(u)\big)$, which forces $uM = 0$ for every $u$ when $k \neq 1$. A gate that vanishes at 0 makes every unit of order $t^2$ near the origin, so both blocks have derivative 0 there, while $xM$ has derivative $M$. A $\tanh$ block is bounded, and $xM$ is not. The condition $k \neq 1$ is needed: a linear block $x \mapsto xA$, whose activation is the identity, absorbs any skip, since $xM + xA = x(M + A)$.

## 3.3 Attention blocks

An *attention block* with parameters $\theta = (W_{Q,i}, W_{K,i}, W_{V,i}, W_{O,i})_{i=1}^H$ maps $h \in \mathbb{R}^{T \times d_{\text{model}}}$, whose rows are the $T \geq 2$ tokens, to

$$B_\theta(h) = \sum_{i=1}^{H} P_{\Gamma_i}(h)\, h\, W_{V,i} W_{O,i}, \qquad P_\Gamma(h) = \operatorname{softmax}\big(h\,\Gamma h^\top\big), \qquad \Gamma_i = W_{Q,i} W_{K,i}^\top / \sqrt{d_h}, \tag{4}$$

where $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$, with $d_h$ the head dimension. The softmax acts on each row of the $T \times T$ matrix of scores, with or without a causal mask, so each row of $P_\Gamma(h)$ is a probability vector. The read of the block holds the matrices $W_{Q,i}$, $W_{K,i}$ and $W_{V,i}$, and its write the matrices $W_{O,i}$. Head $i$ has the score matrix $\Gamma_i$ and the value–output product $W_{V,i} W_{O,i}$.

Attention mixes tokens, and a skip keeps each token where it is. Suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all $h$; two inputs show that $M = 0$. If all tokens are equal, mixing changes nothing, so the sums of the value–output products over the heads of the two blocks must differ by $M$. Now let one token be zero, placed last when the mask is causal. Its query is zero, so every head gives it the plain mean of the tokens it sees, and the two blocks differ there by this mean times $M$. The skip adds zero to this token, so this mean times $M$ vanishes for every choice of the other tokens, and $M = 0$.

**Theorem 3.4** (Attention blocks)**.** *Let $\theta$ and $\theta'$ be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix. For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*

*A special case of Theorem A.12, proved in Appendix A.2.*

The same two inputs show that a residual network of one attention block equals no skipless network of one attention block, with any number of heads (Theorem A.12(iii)). The proof also bounds approximate absorption: on some input with rows of norm at most 1, one row of $hM + B_\theta(h) - B_{\theta'}(h)$ has norm at least $\|M\|/3$, with $\|M\|$ the spectral norm (Theorem A.12(ii)). The proof uses only that each row of attention weights sums to 1 and is uniform for a zero query, so it also holds with rotary position embeddings. With three or more tokens, the function of an attention block determines the

score matrices and the value–output products of its heads, up to merging heads that share a score matrix and dropping heads that output zero (Proposition A.14 in Appendix A.2).

# 4 Networks

*Between an embedding and a readout, the picture survives. Without normalization, no residual network of ReLU$^2$, gated or* $\tanh$ *blocks equals a skipless network, and a skipless ReLU or GELU network needs a wide first block. Under RMSNorm, almost no residual network equals a skipless one.*

Recall that a network embeds its input as $h_0 = XE$ and reads out $y = h_L D$, and that its skips add $h_{\ell-1}$ at every block.

**Theorem 4.1** (Networks that never absorb their skips)**.** *Let the residual network have $ED \neq 0$.*
**(i)** *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let the blocks be gated blocks whose gate vanishes at 0 and is differentiable there, or blocks that are positively homogeneous of degree $k > 1$. No skipless network of blocks of the same kind, of any depth and widths, equals the residual network.*
**(ii)** *Without normalization, no skipless network of* $\tanh$ *blocks, of any depth and widths, equals a residual network of* $\tanh$ *blocks.*

*Proof in Appendix A.1.*

With $E = D = I$, this covers compositions of blocks. The reasons are those of Figure 1, now for the whole network. A gated block, and a homogeneous block of degree $k > 1$, has derivative 0 at the origin. So every skipless network of them has Jacobian 0 at the input $X = 0$, while a residual network has Jacobian $ED \neq 0$ there, or $ED/\sqrt{\varepsilon}$ under RMSNorm. This comparison of derivatives needs $k > 1$. A skipless $\tanh$ network is bounded, while a residual one grows like $XED$ along rays.

**ReLU and GELU networks need a wide first block.**

**Theorem 4.2** (ReLU and GELU networks)**.** *Consider networks of ReLU blocks, or of GELU blocks, without normalization. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

*Proof in Appendix A.1.*

For ReLU the proof compares ranks. Near almost every input, a generic residual ReLU network is a linear map of rank $d_{\text{model}}$. No unit of the first block is active at both an input and its negative, so with fewer than $2d_{\text{model}}$ units a skipless network has fewer than $d_{\text{model}}$ active units at one of the two, and some direction of inputs leaves it unchanged there. GELU reduces to ReLU along rays: $|\text{GELU}(tz) - t\,\text{ReLU}(z)| < 0.17$ for all $z$ and all $t > 0$, so as $t \to \infty$ the output of a GELU network at the input $tX$, divided by $t$, tends to the output at $X$ of the ReLU network with the same weights. Two GELU networks that are equal therefore stay equal with ReLU in place of GELU.

Conversely, the construction of Section 3.1 works block by block.

**Proposition 4.3** (Extra units)**.** *Let $\sigma$ be ReLU or GELU. Without normalization, a residual network whose reads all have rank $d_{\text{model}}$ equals a skipless network with $d_{\text{model}}$ more hidden units in every block.*

*Proof in Appendix A.1.*

So $d_{\text{model}}$ extra units per block suffice, and the first block of a skipless network needs at least $2d_{\text{model}}$ units (Theorem 4.2). For the first block, the two bounds meet when the residual first block has $d_{\text{model}}$ units; beyond that, the number of extra units is open.

**RMSNorm: the skip keeps the norm of each token.** A skipless network sees its input only through its first normalization, and $N(ch)$ depends on $c > 0$ only through $\varepsilon/c^2$: with $\varepsilon = 0$ the norm of each token is lost, and with $\varepsilon > 0$ its effect fades like $\varepsilon/c^2$ as $c$ grows.

**Theorem 4.4** (Networks under RMSNorm)**.** *Consider pre-RMSNorm networks, with $\varepsilon \geq 0$ and blocks of any kind and width. For almost every choice of weights, a residual network equals no skipless network, whatever its depth, widths and kinds of blocks.*

*Proof in Appendix A.3.*

Scale an input $X$ by a factor $t \to \infty$. A skipless network with embedding $E'$ depends on $t$ only through $N(tXE')$, so its output converges at rate $1/t^2$ or faster. In a residual network the skips carry $tXE$ to the last normalization, and every block reads, up to $O(1/t)$, the same normalized input and adds a bounded output. These outputs turn each token by an angle of order $1/t$, so the output of the network converges only at rate $1/t$, unless in every token they add up to a multiple of the token itself. For almost every choice of weights, at some input they do not.

# 5 Skipless networks are limits of residual ones

*A skip removes no function in the limit: without normalization, residual networks of the same widths approach every skipless network.*

**Proposition 5.1** (Skipless networks as limits of residual networks)**.** *Let $F$ be a skipless network whose blocks are locally Lipschitz.*
**(i) Without normalization.** *The network $F$ is a limit of residual networks with blocks of the same kinds and widths, uniformly on bounded sets of inputs.*
**(ii) Under pre-RMSNorm with $\varepsilon = 0$.** *The network $F$ is a limit of residual networks with blocks of the same kinds and widths, uniformly on compact sets of inputs at which $F$ is defined.*

*Proof in Appendix A.4.*

**Blocks that outweigh their skips.** The proof scales the reads of block $\ell$ by $t^{1-\ell}$, its writes by $t^\ell$ and the readout by $t^{-L}$. Block $\ell$ then sees the hidden state of $F$, up to $O(1/t)$, and outweighs its skip by a factor $t$, so the residual network computes $F + O(1/t)$ (Remark A.15). The large constants sit in the writes, so no activation sees a large argument, and the construction applies alike to $\tanh$, GELU, ReLU, ReLU$^2$, gated and attention blocks. The weights diverge as $t \to \infty$: the proposition says which functions residual networks reach in the limit, and leaves open how to parametrize them for training.

**Under RMSNorm.** With $\varepsilon = 0$, RMSNorm ignores the size of the hidden states, so only the writes grow. With $\varepsilon > 0$ the construction fails, since on a hidden state of size $t^\ell$ RMSNorm acts as RMSNorm with $\varepsilon/t^{2\ell}$; whether skipless networks are limits of residual ones there is open.

# 6 Related work

**Expressivity of residual networks.** Lin and Jegelka (2018) show that residual ReLU networks with one hidden unit per block are universal approximators, which networks of width $d_{\text{in}}$ without skips are not; they count the identity map as $d_{\text{model}}$ hidden units. For blocks without biases and with at least $d_{\text{model}}$ units, Theorem 3.1(ii) shows that this count is exact for almost every ReLU or GELU block, and Theorem 3.3 that no count works for ReLU$^2$, gated and $\tanh$ blocks. With biases, as in their blocks, the count stays exact for almost every ReLU or GELU block with at least $d_{\text{model}}$ units (Theorem C.2 in Appendix C). Hardt and Ma (2017) show that deep linear residual networks have no spurious local optima, and that residual ReLU networks can fit any finite sample once they have more parameters than samples. These results concern approximation, optimization or finite samples; ours concern exact equality of functions, block by block and activation by activation.

**Identifiability.** For $\tanh$ networks, the function determines the weights up to sign flips and permutations of hidden units (Fefferman, 1994; Sussmann, 1992), and similar statements hold for other activations (Vlačić and Bölcskei, 2022) and for ReLU networks (Petzka et al., 2020; Phuong and Lampert, 2020; Rolnick and Kording, 2020). Our proofs identify the units of a block from its function in the same spirit, and then ask what is left for the linear term of the skip.

**Training without skips.** Skipless networks train with careful initialization or activation shaping (Martens et al., 2021; Zhang et al., 2022), and skipless transformers with modified attention (He et al., 2023; Noci et al., 2023) or simplified blocks (He and Hofmann, 2024). Ji et al. (2026) train a

skipless ViT-B that outperforms the residual one. Our results say when a skipless block or network can compute exactly what a residual one computes, and with how many units.

# 7 Conclusion

Can a block absorb its skip? Without biases, it depends on the activation. At the same width, almost no block absorbs its skip. Almost every ReLU or GELU block needs exactly $d_{\mathrm{model}}$ more units, and ReLU$^2$, gated, $\tanh$ and attention blocks cannot absorb the skip at any width. Between an embedding and a readout and without normalization, no skipless network of ReLU$^2$, gated or $\tanh$ blocks equals a residual one, and a skipless ReLU or GELU network needs a first block of at least $2d_{\mathrm{model}}$ units to equal almost any residual one. Under RMSNorm, with any $\varepsilon \geq 0$, no width suffices for almost every residual network. So, without normalization or biases, a trained residual ReLU or GELU network whose reads have rank $d_{\mathrm{model}}$ can be rewritten exactly without skips with $d_{\mathrm{model}}$ more units per block, and a ReLU$^2$, gated or $\tanh$ network with $ED \neq 0$ cannot be rewritten at any width. With biases, ReLU$^2$ and gated networks can be rewritten too (Appendix C).

**Open questions.** The unknown cells and the conditions of Table 1 leave three questions, and a fourth lies beyond the table.

(1) Networks of attention blocks at depth, without normalization. At the same widths, a companion paper answers this by counting functional dimensions, the numbers of directions in which the parameters move the function (Karbevski and Mijoski, 2026); with more heads it is open.
(2) How many extra units ReLU and GELU networks need, without normalization: $d_{\mathrm{model}}$ more per block suffice, and the first block needs at least $2d_{\mathrm{model}}$ units.
(3) Skipless networks as limits of residual networks under RMSNorm with $\varepsilon > 0$.
(4) Approximate absorption: how many units a skip is worth at a given error.

# Use of AI

The conception of the study is the authors' own. One author obtained their results, proofs and experiments working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs and to write the code of the experiments, which the authors validated.

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# A Proofs

## A.1 MLP blocks and networks (Sections 3 and 4)

We recall Theorems 3.1 and 3.3 and Proposition 3.2 from Section 3 and Theorems 4.1 and 4.2 and Proposition 4.3 from Section 4, for the question (2). Below we write $B$ for $B_\theta$, and $B'$ for $B_{\theta'}$ with $\theta' = (W'_{\text{read}}, W'_{\text{write}})$.

**Theorem 3.1** (ReLU and GELU blocks)**.** *Let $\sigma$ be ReLU or GELU, and fix a matrix $M \neq 0$.*
**(i)** *If $B_\theta$ is nondegenerate, some block $B_{\theta'}$ of the same width satisfies (2) if and only if some set $S$ of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$. For generic $\theta$, no block of the same width does.*
**(ii)** *If $M$ is invertible and $W_{\text{read}}$ has rank $d_{\text{model}}$, some block $B_{\theta'}$ with $d_{\text{model}}$ more hidden units satisfies (2), and for generic $\theta$ none with fewer does.*

**Proposition 3.2** (Networks of one block)**.** *Let $\sigma$ be ReLU or GELU. For almost every choice of weights, a residual network of one block equals no skipless network of one block of the same width.*

**Theorem 3.3** (No width suffices)**.** *Let $B_\theta$ be a* $\tanh$*, ReLU$^2$, ReGLU, SwiGLU or GeGLU block, or more generally a block that is positively homogeneous of degree $k \neq 1$, or a gated block whose gate vanishes at* $0$ *and is differentiable there. For every $M \neq 0$, no block $B_{\theta'}$ of the same kind and of any width satisfies* (2)*.*

**Theorem 4.1** (Networks that never absorb their skips)**.** *Let the residual network have $ED \neq 0$.*
**(i)** *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let the blocks be gated blocks whose gate vanishes at* $0$ *and is differentiable there, or blocks that are positively homogeneous of degree $k > 1$. No skipless network of blocks of the same kind, of any depth and widths, equals the residual network.*
**(ii)** *Without normalization, no skipless network of* $\tanh$ *blocks, of any depth and widths, equals a residual network of* $\tanh$ *blocks.*

**Theorem 4.2** (ReLU and GELU networks)**.** *Consider networks of ReLU blocks, or of GELU blocks, without normalization. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

**Proposition 4.3** (Extra units)**.** *Let $\sigma$ be ReLU or GELU. Without normalization, a residual network whose reads all have rank $d_{\text{model}}$ equals a skipless network with $d_{\text{model}}$ more hidden units in every block.*

**Overview.** Theorem 3.3 compares orders. Along rays from the origin a linear skip scales like $\lambda$ and a block that is homogeneous of degree $k \neq 1$ like $\lambda^k$, and near the origin a gated block is of second order, so the two sides of the absorption equation cannot agree (Propositions A.2 and A.3); comparing Jacobians at the origin extends this to networks of any depth, Theorem 4.1 (Corollary A.4). Theorem 3.1(i) is the substantial case. The absorption equation splits into its odd and even parts in $x$. The even part determines the hidden units of the absorbing block up to sign, and for ReLU up to a positive rescaling, through the kinks for ReLU (Theorem A.5) and through linear independence of ridge functions for GELU (Lemma A.6 and Theorem A.7). The odd part of ReLU and GELU is exactly $z/2$, so the odd part of the equation is linear and reads off the set $S$ of units whose signs flip; conversely, flipping the signs of the units in $S$ absorbs the skip whenever the condition holds. The generic clause holds because the condition is a finite union of algebraic sets of positive codimension (Corollary A.10), and Remark A.9 covers singular $M$. Theorem 3.1(ii) counts the extra units that the matching requires (Proposition A.11). Proposition 4.3 applies Theorem 3.1(ii) block by block. Theorem 4.2 compares the ranks of the linear pieces of a ReLU network at an input and at its negative, and reduces GELU to ReLU along rays.

The arguments show that the function of a block determines its parameters, in the manner used for $\tanh$ networks by Sussmann (1992) and Fefferman (1994) and for ReLU networks by Phuong and Lampert (2020), Petzka et al. (2020) and Rolnick and Kording (2020): kink alignment for ReLU, ridge-function independence for GELU. For two-layer ReLU networks with a single output, Petzka et al. (2020, Section 3) already give the corresponding sign flips, show that irreducible networks computing the same function differ only by these flips, permutations and positive rescalings (their Theorem 1), and state the matching criterion for removing a linear term. Lemma A.1 reduces any invertible skip to the identity skip.

**Setting.** Throughout, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, $\sigma$ acts coordinatewise, $w_i$ is column $i$ of $W_{\text{read}}$ and $\hat{w}_i$ is row $i$ of $W_{\text{write}}$, and inputs $x$ are row vectors. For $S \subseteq \{1, \dots, m\}$, $W_{\text{read}}[:, S]$ is the submatrix of columns in $S$, $W_{\text{write}}[S, :]$ the submatrix of rows in $S$, and $\Pi_S$ the diagonal projector with $(\Pi_S)_{ii} = 1$ exactly when $i \in S$, so that $W_{\text{read}} \Pi_S W_{\text{write}} = W_{\text{read}}[:, S] W_{\text{write}}[S, :]$. A family of vectors is *pairwise non-collinear* if no one of them is a scalar multiple of another, and a block is *nondegenerate* if the columns of $W_{\text{read}}$ are nonzero and pairwise non-collinear and the rows of $W_{\text{write}}$ are nonzero. The question is whether there is a block $B'(x) = \sigma(xW'_{\text{read}})W'_{\text{write}}$ of the same width, with $W'_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W'_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, such that

$$x + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{5}$$

**Lemma A.1** (Invertible reduction)**.** *Let $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be invertible. Then*

$$xM + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \quad \textit{for all } x$$

*if and only if*

$$x + \sigma(xW_{\text{read}})W_{\text{write}}M^{-1} = \sigma(xW'_{\text{read}})W'_{\text{write}}M^{-1} \quad \textit{for all } x.$$

*Proof.* Right-multiply by $M^{-1}$, which is invertible. □

Any invertible linear skip therefore reduces to the identity skip.

**Proposition A.2** (Homogeneous blocks)**.** *Let $B$ and $B'$ be blocks that are positively homogeneous of degree $k \neq 1$, of any widths, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some $x$.*

Examples are $\mathrm{ReLU}^2$ and ReGLU blocks without biases, both of degree 2.

*Proof. Outline.* We compare the two sides along rays through the origin: a term of degree one and a term of degree $k \neq 1$ cannot agree at every scale.

Suppose it holds for all $x$ and evaluate at $\lambda x$ with $\lambda > 0$: $\lambda xM + \lambda^k B(x) = \lambda^k B'(x)$. Dividing by $\lambda$ gives $xM = \lambda^{k-1}\big(B'(x) - B(x)\big)$, and $\lambda^{k-1} \to 0$ as $\lambda \to 0$ if $k > 1$ and as $\lambda \to \infty$ if $k < 1$. So $xM = 0$ for all $x$, which contradicts $M \neq 0$. □

**Proposition A.3** (Gated blocks)**.** *Let $B$ and $B'$ be gated blocks without biases, of any widths, whose gates are differentiable at 0 and vanish there, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some $x$.*

Examples are SwiGLU and GeGLU blocks, which Shazeer (2020) defines without biases.

*Proof. Outline.* Near the origin a gated block is of second order in $x$, while $xM$ is of first order.

A unit of a gated block computes $g(u \cdot x)\,(v \cdot x)$. Since $g(0) = 0$ and $g$ is differentiable at 0, $|g(u \cdot x)| \leq c\,\|x\|$ for small $\|x\|$, so the unit is $O(\|x\|^2)$, and so are $B$ and $B'$. If $xM + B(x) = B'(x)$ for all $x$, then $xM = O(\|x\|^2)$ as $x \to 0$. Along a ray $x = tz$ this reads $t\,zM = O(t^2)$, so $zM = 0$ for every $z$, which contradicts $M \neq 0$. □

**Corollary A.4** (Any depth)**.** *Let every block be as in Proposition A.2 with $k > 1$ or as in Proposition A.3, of any widths. Let $F$ be a composition of residual blocks $h \mapsto h + B_\ell(h)$ and $H$ a composition of one or more blocks without skips. Then $F \neq H$. With embeddings and readouts, the residual network $X \mapsto F(XE)D$ differs from the skipless network $X \mapsto H(XE')D'$ whenever $ED \neq 0$, also with pre-RMSNorm and $\varepsilon > 0$. Without normalization the same holds for $\tanh$ blocks.*

*Proof.* Each such block maps 0 to 0 and has Jacobian 0 there, because it is $O(\|x\|^k)$ with $k > 1$, respectively $O(\|x\|^2)$. By the chain rule $F$ has Jacobian $I$ at the origin and $H$ has Jacobian 0; with embeddings and readouts these become $ED$ and 0. Under pre-RMSNorm with $\varepsilon > 0$, $N(0) = 0$ and $N$ has Jacobian $I/\sqrt{\varepsilon}$ at 0, so every block, read through $N$, still has Jacobian 0 there, and the residual and the skipless network have Jacobians $ED/\sqrt{\varepsilon}$ and 0 at the origin. For $\tanh$ blocks without normalization, $H$ and $F(h) - h$ are bounded, so $F(XE)D - H(XE')D'$ differs from $XED$ by a bounded function and does not vanish when $ED \neq 0$. □

**Parity.** The remaining activations have an odd part that is exactly linear. For ReLU, $\mathrm{ReLU}(z) = |z|/2 + z/2$. For GELU, $\mathrm{GELU}(z) = z\Phi(z)$ with $\Phi$ the standard normal distribution function, and since $\Phi(-z) = 1 - \Phi(z)$,

$$\mathrm{GELU}(z) = \Psi(z) + z/2, \qquad \Psi(z) := z\big(\Phi(z) - \tfrac{1}{2}\big), \qquad \Psi(-z) = \Psi(z).$$

The even parts, $|z|/2$ and $\Psi$, match the hidden units of the two blocks; the odd part, $z/2$, turns a sign flip into a linear term and gives the absorption condition.

**Theorem A.5** (Identity absorption for ReLU)**.** *Let $\sigma = \mathrm{ReLU}$, $d_{\mathrm{model}} \geq 2$, and let the block be nondegenerate. Then (5) has a solution if and only if*

$$W_{\mathrm{read}}[:, S]\, W_{\mathrm{write}}[S, :] = -I_{d_{\mathrm{model}}} \quad \textit{for some } S. \tag{6}$$

*Proof. Outline.* Sufficiency is a construction: flipping the signs of the hidden units indexed by $S$ produces the missing identity term. For necessity we split (5) into its odd and even parts in $x$, use the kinks of the even part to match the hidden units of the two blocks one to one, and read the index set $S$ off the signs of the matching.

*Sufficiency.* Let $F = I_m - 2\Pi_S$, a diagonal sign matrix, and set $W'_{\mathrm{read}} = W_{\mathrm{read}}F$, $W'_{\mathrm{write}} = W_{\mathrm{write}}$. Since $|xW_{\mathrm{read}}F| = |xW_{\mathrm{read}}|$ coordinatewise,

$$\begin{aligned} \mathrm{ReLU}(xW'_{\mathrm{read}})W'_{\mathrm{write}} &= \tfrac{1}{2}\big(xW_{\mathrm{read}}F + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}}, \\ x + \mathrm{ReLU}(xW_{\mathrm{read}})W_{\mathrm{write}} &= \tfrac{1}{2}\big(xW_{\mathrm{read}} + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}} + x, \end{aligned}$$

and the two agree if and only if $\frac{1}{2}xW_{\mathrm{read}}(F - I_m)W_{\mathrm{write}} = x$ for all $x$, that is $-W_{\mathrm{read}}\Pi_S W_{\mathrm{write}} = I_{d_{\mathrm{model}}}$, which is (6).

*Necessity.* Suppose (5) holds and split both sides into odd and even parts in $x$:

$$W'_{\mathrm{read}}W'_{\mathrm{write}} = W_{\mathrm{read}}W_{\mathrm{write}} + 2I_{d_{\mathrm{model}}}, \qquad |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}} = |xW_{\mathrm{read}}|\, W_{\mathrm{write}} \ \text{ for all } x. \tag{7}$$

*Step 1, alignment of kinks.* The map $x \mapsto |xW_{\mathrm{read}}|\, W_{\mathrm{write}}$ fails to be differentiable exactly on the union of the hyperplanes $\{xw_i = 0\}$; these are $m$ distinct hyperplanes, each carrying a genuine kink since $\hat{w}_i \neq 0$. By the second identity in (7) the map $x \mapsto |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}}$ has the same non-differentiability locus. Each of the $m$ kink hyperplanes $\{x : xw_i = 0\}$ of the target needs at least one unit of $B'$ whose kink lies on it, and $B'$ has only $m$ units; so each hyperplane gets exactly one unit, no two units share a hyperplane, and no kinks cancel. After relabeling, $W'_{\mathrm{read}} = W_{\mathrm{read}}C$ with $C = \mathrm{diag}(c_1, \ldots, c_m)$, all $c_i \neq 0$. *Step 2, coefficients.* Substituting, $|xW_{\mathrm{read}}C| = |xW_{\mathrm{read}}|\,|C|$, so $|xW_{\mathrm{read}}|\,|C|\,W'_{\mathrm{write}} = |xW_{\mathrm{read}}|\,W_{\mathrm{write}}$ for all $x$. For $d_{\mathrm{model}} \geq 2$ and pairwise non-collinear columns, the functions $|xw_i|$ are linearly independent modulo linear functions, because their kink hyperplanes are distinct; hence $|C|\,W'_{\mathrm{write}} = W_{\mathrm{write}}$ and $W'_{\mathrm{write}} = |C|^{-1}W_{\mathrm{write}}$. *Step 3, signs.* The first identity in (7) becomes $W_{\mathrm{read}}C|C|^{-1}W_{\mathrm{write}} = W_{\mathrm{read}}W_{\mathrm{write}} + 2I_{d_{\mathrm{model}}}$. With $\Sigma = C|C|^{-1} = \mathrm{diag}(\mathrm{sign}\, c_i)$ this reads $W_{\mathrm{read}}(\Sigma - I_m)W_{\mathrm{write}} = 2I_{d_{\mathrm{model}}}$; the diagonal of $\Sigma - I_m$ has entries $0$ or $-2$, so with $S = \{i : c_i < 0\}$ we get $W_{\mathrm{read}}\Pi_S W_{\mathrm{write}} = -I_{d_{\mathrm{model}}}$, and $|S| \geq d_{\mathrm{model}}$ by the rank of $I_{d_{\mathrm{model}}}$. □

The GELU case needs one classical fact about ridge functions, whose short proof we include.

**Lemma A.6** (Vanishing sums of ridge functions)**.** *Let $a_1, \ldots, a_r \in \mathbb{R}^{d_{\mathrm{model}}}$ be nonzero and pairwise non-collinear, and let $g_1, \ldots, g_r : \mathbb{R} \to \mathbb{R}^k$ be $r-1$ times continuously differentiable with $\sum_i g_i(a_i \cdot x) = 0$ for all $x \in \mathbb{R}^{d_{\mathrm{model}}}$. Then every $g_i$ is a polynomial of degree at most $r-2$.*

*Proof.* Fix $i$. For each $j \neq i$ choose $u_j \in \mathbb{R}^{d_{\mathrm{model}}}$ with $u_j \cdot a_j = 0$ and $u_j \cdot a_i \neq 0$, which exists because $a_i$ is not a multiple of $a_j$. The directional derivative $\partial_{u_j}$ annihilates every function of $a_j \cdot x$ and maps a function of $a_k \cdot x$ to a function of $a_k \cdot x$. Applying the product of the commuting operators $\partial_{u_j}$, $j \neq i$, to the sum therefore leaves $\prod_{j \neq i}(u_j \cdot a_i)\, g_i^{(r-1)}(a_i \cdot x) = 0$. Since $a_i \neq 0$, $a_i \cdot x$ takes every real value, so $g_i^{(r-1)} = 0$. □

**Theorem A.7** (Identity absorption for GELU)**.** *Let $\sigma = \mathrm{GELU}$ and let the block be nondegenerate. Then (5) has a solution if and only if (6) holds.*

*Proof. Outline.* The proof follows that of Theorem A.5. Linear independence of ridge functions replaces the kink argument, and one extra step shows that matched hidden units have the same scale.

*Sufficiency.* With $F = I_m - 2\Pi_S$, $W'_{\text{read}} = W_{\text{read}}F$ and $W'_{\text{write}} = W_{\text{write}}$, evenness of $\Psi$ gives $\Psi(xW_{\text{read}}F) = \Psi(xW_{\text{read}})$, so

$$\begin{aligned}\text{GELU}(xW'_{\text{read}})W'_{\text{write}} &= \Psi(xW_{\text{read}})W_{\text{write}} + \tfrac{1}{2}xW_{\text{read}}FW_{\text{write}}\\ &= \text{GELU}(xW_{\text{read}})W_{\text{write}} - xW_{\text{read}}\Pi_S W_{\text{write}}\\ &= \text{GELU}(xW_{\text{read}})W_{\text{write}} + x.\end{aligned}$$

*Necessity.* Splitting (5) into even and odd parts gives

$$\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}} \ \text{ for all } x, \qquad W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}. \tag{8}$$

*Step 1, alignment of directions.* $\Psi$ is analytic and not a polynomial. Zero columns of $W'_{\text{read}}$ contribute $\Psi(0) = 0$; group the other columns of $W'_{\text{read}}$ and the columns of $W_{\text{read}}$ into classes of collinear vectors. In the first identity of (8) the terms of each class form one ridge function in the direction of the class, and these ridge functions sum to zero, so each is a polynomial by Lemma A.6. The class of a column $w_i$ of $W_{\text{read}}$ contains no other column of $W_{\text{read}}$, so if it contained no column of $W'_{\text{read}}$ its ridge function would be $-\Psi(xw_i)\,\hat{w}_i$, which is not a polynomial because $\hat{w}_i \neq 0$. Hence each of the $m$ classes of the columns of $W_{\text{read}}$ contains a nonzero column of $W'_{\text{read}}$, and since $W'_{\text{read}}$ has only $m$ columns, after relabeling $W'_{\text{read}} = W_{\text{read}}C$ with $C = \text{diag}(c_i)$, $c_i \neq 0$. *Step 2, unit scale.* Collecting the terms of direction $w_i$ on both sides gives, for each $i$, a ridge function $g_i(xw_i) := \Psi(|c_i|\,xw_i)\,(W'_{\text{write}})_{i,:} - \Psi(xw_i)\,(W_{\text{write}})_{i,:}$ with $\sum_i g_i(xw_i) = 0$. By Lemma A.6 each $g_i$ is a polynomial. For $t \to +\infty$, $\Psi(t) = t/2 - t\,\bar{\Phi}(t)$ with $\bar{\Phi}(t) = 1 - \Phi(t)$ exponentially small, so $g_i(t) = \frac{t}{2}\big(|c_i|(W'_{\text{write}})_{i,:} - (W_{\text{write}})_{i,:}\big) + o(1)$; a polynomial with this behavior is linear, and it is even because $\Psi$ is even, so it is constant, and letting $t \to 0$ shows the constant is zero. Hence $|c_i|(W'_{\text{write}})_{i,:} = (W_{\text{write}})_{i,:}$ and the exponentially small remainder $|c_i|\,t\,\bar{\Phi}(|c_i|t)\,(W'_{\text{write}})_{i,:} - t\,\bar{\Phi}(t)\,(W_{\text{write}})_{i,:}$ must vanish, which with $(W_{\text{write}})_{i,:} \neq 0$ forces $\bar{\Phi}(|c_i|t) = \bar{\Phi}(t)$ for all $t > 0$, so $|c_i| = 1$. Therefore $|C| = I_m$ and $W'_{\text{write}} = W_{\text{write}}$. *Step 3, signs.* Identical to the ReLU case: the second identity in (8) becomes $W_{\text{read}}(C - I_m)W_{\text{write}} = 2I_{d_{\text{model}}}$ with $C = \text{diag}(\pm 1)$, so $S = \{i : c_i = -1\}$ satisfies (6) and $|S| \geq d_{\text{model}}$. □

*Remark* A.8. Any activation whose even part $\psi$ is analytic and not a polynomial matches the hidden units in the same way: Steps 1 and 2 of the proof of Theorem A.7 apply with $\psi$ in place of $\Psi$ (zero columns of $W'_{\text{read}}$ now contribute a constant, which is a polynomial), except that the unit scale in Step 2 follows from a general fact: if $\sum_k a_k\psi(\alpha_k t)$ is a polynomial for distinct $\alpha_k > 0$, then every $a_k = 0$. Indeed, the Taylor coefficients $\psi_n \sum_k a_k\alpha_k^n$ vanish for all large $n$, infinitely many $\psi_n$ are nonzero, and a generalized Vandermonde matrix with distinct positive nodes is invertible. Linear independence of the ridge functions alone does not suffice: for $\sigma(z) = z/2 + |z|^3$ the even part is homogeneous, so a unit can be rescaled together with its output row, and the unit scale is not forced. The odd part $o$ of $\sigma$ enters only through the flips: with the units matched, both directions reduce to $x = -2\sum_{i\in S} o(xw_i)\,\hat{w}_i$, with $S$ the flipped units. For $o(z) = z/2$, that is $\sigma(z) - \sigma(-z) = z$, as for SiLU, whose even part is $\frac{z}{2}\tanh\frac{z}{2}$, this is (6), and $o(z) = \alpha z$ with $\alpha \neq 0$ only rescales it; a polynomial $o$ of higher degree adds the condition that the units in $S$ cancel its terms of degree three and higher, and a smooth $o$ that is not a polynomial admits no $S$ (Lemma A.6), so the skip is never absorbed at the same width.

*Remark* A.9 (Singular linear skips). Nothing in the proofs of Theorems A.5 and A.7 uses that the linear term is the identity. With $xM$ in place of $x$ on the left of (5), the odd part of the equation reads $W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2M$, the even part is unchanged, and the sign step yields $W_{\text{read}}\Pi_S W_{\text{write}} = -M$. So for any $M \neq 0$, invertible or not, absorption holds if and only if $W_{\text{read}}[:, S]W_{\text{write}}[S, :] = -M$ for some $S$, with $|S| \geq \text{rank}\,M$, which is a measure-zero condition on the weights for each fixed $M$. This gives Theorem 3.1(i) for singular $M$. The proofs also hold verbatim for rows $x \in \mathbb{R}^{1\times p}$ with $p \geq 2$, reads in $\mathbb{R}^{p\times m}$, writes in $\mathbb{R}^{m\times q}$ and $M \in \mathbb{R}^{p\times q}$: the kinks and the ridge functions live in $\mathbb{R}^p$, and the sign step gives $W_{\text{read}}[:, S]\,W_{\text{write}}[S, :] = -M$ for any shape. This covers networks of one block, with the reads $EW_{\text{read}}$, the writes $W_{\text{write}}D$ and $M = ED$, so that $p = d_{\text{in}}$ and $q = d_{\text{out}}$. For generic $E$, $W_{\text{read}}$, $W_{\text{write}}$ and $D$, these reads and writes are

nondegenerate, and $EW_{\text{read}}[:, S]\, W_{\text{write}}[S, :]\, D \neq -ED$ for every $S$, since the two sides differ at $W_{\text{write}} = 0$ when $ED \neq 0$.

**Corollary A.10** (Generic disjointness for one block)**.** *Let $\sigma$ be ReLU or GELU and $d_{\text{model}} \geq 2$. For generic weights,* (5) *has no solution.*

*Proof. Outline.* The condition of Theorems A.5 and A.7 is a finite union of algebraic sets of positive codimension, and such a union has measure zero.

For each of the finitely many $S$, (6) is $d_{\text{model}}^2$ polynomial equations in the weights and defines a set of measure zero; the hypotheses of Theorems A.5 and A.7 hold outside another set of measure zero. So a generic residual block is not a block of the same width without a skip, and, by the same argument with the two sides of (5) exchanged, a generic block without a skip is not a residual block of the same width: neither class contains the other. □

**Proposition A.11** ($d_{\text{model}}$ more units)**.** *Let $\sigma$ be ReLU or GELU, let $d_{\text{model}} \geq 2$, and let $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be invertible.*
**(i)** *If $W_{\text{read}}$ has rank $d_{\text{model}}$, a block $B'$ of width $m + d_{\text{model}}$ satisfies $xM + B(x) = B'(x)$ for all $x$.*
**(ii)** *For generic weights no block of width less than $m + d_{\text{model}}$ does.*

*Proof. Outline.* Two units with opposite reads produce a linear term of rank one, so $d_{\text{model}}$ extra units produce $xM$. Conversely, the matching of the units in Theorems A.5 and A.7 shows that every extra unit adds at most one such term.

(i) As $\sigma(z) - \sigma(-z) = z$, a column $w$ and rows $a$, $b$ give $\sigma(xw)(a + b) + \sigma(-xw)(-b) = \sigma(xw)\, a + (xw)\, b$. Choose $d_{\text{model}}$ units whose columns $w_{i_1}, \ldots, w_{i_{d_{\text{model}}}}$ form an invertible matrix $W_K$, and let $b_1, \ldots, b_{d_{\text{model}}}$ be the rows of $W_K^{-1} M$. Adding $b_k$ to the row $\hat{w}_{i_k}$, and a unit with column $-w_{i_k}$ and row $-b_k$, for each $k$, gives a block of width $m + d_{\text{model}}$ that computes $\sigma(xW_{\text{read}})W_{\text{write}} + xW_K W_K^{-1} M$.

(ii) Outside a set of measure zero the block is nondegenerate, and for each of the finitely many sets $S$ of units the determinant of $M + W_{\text{read}}[:, S] W_{\text{write}}[S, :]$ is nonzero, since it is a polynomial in the weights equal to $\det M \neq 0$ at $W_{\text{write}} = 0$. Let a block $B'$ of width $m'$, with columns $w'_j$ and rows $\hat{w}'_j$, satisfy $xM + B(x) = B'(x)$. Its odd part gives $W'_{\text{read}} W'_{\text{write}} = W_{\text{read}} W_{\text{write}} + 2M$, and its even part $|xW'_{\text{read}}|\, W'_{\text{write}} = |xW_{\text{read}}|\, W_{\text{write}}$ for ReLU and $\Psi(xW'_{\text{read}}) W'_{\text{write}} = \Psi(xW_{\text{read}}) W_{\text{write}}$ for GELU. Group the nonzero columns $w'_j$ and the columns $w_i$ into classes of collinear vectors, write $w'_j = s_j u$ with a unit vector $u$ of the class, $u = w_i / \|w_i\|$ in the class of $w_i$, and let $P_u = \|w_i\| \hat{w}_i$ there and $P_u = 0$ in a class without a column of $W_{\text{read}}$. The even part then forces, class by class, $\sum_j |s_j|\, \hat{w}'_j = P_u$ for ReLU, since the functions $|xu|$ of distinct classes are linearly independent (Step 2 of Theorem A.5), and for GELU, by Lemma A.6 and the independence of the functions $\Psi(\alpha t)$ for distinct $\alpha > 0$ modulo polynomials (Remark A.8), $\sum_{|s_j| = \alpha} \hat{w}'_j = \hat{w}_i$ if $w_i$ lies in the class and $\alpha = \|w_i\|$, and 0 otherwise. With $Q_u = \sum_j s_j \hat{w}'_j$ the odd part reads $\sum_u u\, (Q_u - P_u) = 2M$. Call a class mixed if it contains $w'_j$ of both signs, for GELU of both signs and equal $|s_j|$. A class that is not mixed has $Q_u = P_u$ or $Q_u = -P_u$; let $S$ collect the units $i$ whose class is not mixed and has $Q_u = -P_u \neq 0$. Then

$$2\big(M + W_{\text{read}}[:, S] W_{\text{write}}[S, :]\big) = \sum_{u \text{ mixed}} u\, (Q_u - P_u).$$

Every class of a column $w_i$ contains some $w'_j$, since $P_u \neq 0$, and every mixed class contains at least two, so $m' \geq m$ plus the number of mixed classes. If $m' < m + d_{\text{model}}$, the right side has rank less than $d_{\text{model}}$, so the left side is singular, which the weights exclude. □

*Proof of Theorems 3.3 and 4.1.* For homogeneous and gated blocks, Theorem 3.3 is Propositions A.2 and A.3, and Theorem 4.1(i) is Corollary A.4. For $\tanh$ blocks, $B_\theta$ and $B_{\theta'}$ are bounded and $xM$ is not, which gives Theorem 3.3, and Theorem 4.1(ii) is again Corollary A.4. □

*Proof of Theorem 3.1.* Part (i) is Theorems A.5 and A.7, with Lemma A.1 for invertible $M$ and Remark A.9 for singular $M$; for each $M$ the condition holds only on a set of weights of measure zero (Corollary A.10 and Remark A.9). Part (ii) is Proposition A.11. □

*Proof of Proposition 3.2.* A network of one block computes $X \mapsto XED + f(XEW_{\text{read}})\, W_{\text{write}}D$, and a skipless network of one block of the same width computes $X \mapsto f(XE'W'_{\text{read}})\, W'_{\text{write}}D'$. Both act on each row of $X$ separately. By the rectangular case of Remark A.9, with the reads $EW_{\text{read}}$, the writes $W_{\text{write}}D$ and $M = ED$, they are equal only if $EW_{\text{read}}[:, S]\, W_{\text{write}}[S, :]\, D = -ED$ for some set $S$ of units, or if the reads and writes are degenerate, and for generic weights neither happens. □

*Proof of Theorem 4.2.* MLP blocks act on each token separately, so we compare the networks on rows $x \in \mathbb{R}^{1\times d_{\text{in}}}$. Outside a set of weights of measure zero, $E$ and $D$ have rank $d_{\text{model}}$, no read of a unit vanishes, and $I + W^{\ell}_{\text{read}}[:, S]\, W^{\ell}_{\text{write}}[S, :]$ is invertible for every block $\ell$ and every set $S$ of its units, since each determinant is a polynomial in the weights that equals 1 when the writes vanish. On an open set of inputs on which no unit of the residual network switches, with active sets $S_\ell$, the network is the linear map $x \mapsto xEMD$ with $M = \prod_\ell \big(I + W^{\ell}_{\text{read}}[:, S_\ell]\, W^{\ell}_{\text{write}}[S_\ell, :]\big)$ invertible, of rank $d_{\text{model}}$. The inputs at which a unit switches lie in finitely many hyperplanes, by induction over the blocks: off the hyperplanes of blocks $1, \dots, \ell - 1$, the active sets of these blocks take finitely many values, and for each of them the pre-activation of a unit of block $\ell$ is $x$ times a vector $EM'w$, with $M'$ invertible and $w \neq 0$ the read of the unit. So the inputs near which the network is linear of rank $d_{\text{model}}$ form an open dense set $\Omega$.

Let a skipless network $F'$ have embedding $E'$ and a first block with reads $w'_1, \dots, w'_{m'}$, $m' < 2d_{\text{model}}$, and set $a_i = E'w'_i$. Take $x_0$ with $x_0, -x_0 \in \Omega$ and $x_0 a_i \neq 0$ whenever $a_i \neq 0$. The units active near $x_0$, those with $x_0 a_i > 0$, and those active near $-x_0$ are disjoint, so one of the two sets, say the first after replacing $x_0$ by $-x_0$ if needed, has fewer than $d_{\text{model}}$ units. Near $x_0$ the other units of the first block output zero, so $F'$ depends on $x$ only through the numbers $xa_i$ of the active units, and $F'(x_0 + u) = F'(x_0)$ for every small $u$ in a subspace of dimension greater than $d_{\text{in}} - d_{\text{model}}$. The residual network $F$ changes by $uEMD$, which vanishes only on a subspace of dimension $d_{\text{in}} - d_{\text{model}}$. So some small $u$ changes $F$ and leaves $F'$ unchanged, and $F' \neq F$.

*GELU.* As $\text{GELU}(y) - \text{ReLU}(y) = -|y|\, \Phi(-|y|)$, with $\Phi$ the standard normal distribution function, we have $|\text{GELU}(ty)/t - \text{ReLU}(y)| \leq \kappa/t$ for all $y \in \mathbb{R}$ and $t > 0$, where $\kappa = \max_{u\geq 0} u\, \Phi(-u) < 0.17$. Given a network of GELU blocks without normalization, let $h_\ell(t)$ be its hidden states at the input $tx$, and $h^{\text{R}}_\ell$ those of the ReLU network with the same weights at the input $x$. Then $h_0(t)/t = h^{\text{R}}_0$. If $h_{\ell-1}(t)/t = h^{\text{R}}_{\ell-1} + O(1/t)$, the output of block $\ell$, divided by $t$, is $\text{GELU}\big(h_{\ell-1}(t)W^{\ell}_{\text{read}}\big)W^{\ell}_{\text{write}}/t = \text{ReLU}\big(h^{\text{R}}_{\ell-1}W^{\ell}_{\text{read}}\big)W^{\ell}_{\text{write}} + O(1/t)$, by the bound and since ReLU is 1-Lipschitz, so $h_\ell(t)/t = h^{\text{R}}_\ell + O(1/t)$, with or without the skip. So the output of the GELU network at $tx$, divided by $t$, tends to the output of the ReLU network at $x$. If a residual GELU network equals a skipless one, the two ReLU networks with the same weights are therefore equal. By the case of ReLU, the weights of the residual network for which this happens with a skipless network whose first block has fewer than $2d_{\text{model}}$ units form a set of measure zero. □

*Proof of Proposition 4.3.* Theorem 3.1(ii) with $M = I$ turns each residual block, whose read has rank $d_{\text{model}}$, into a block without a skip with $d_{\text{model}}$ more hidden units. Composing these blocks, with the same embedding and readout, gives a skipless network that equals the residual one. □

## A.2 No absorption in attention (Theorem 3.4, Proposition A.14)

Theorem 3.4 collects what the main text uses of the following theorem, for the attention blocks $B_\theta$ of (4).

**Theorem A.12** (Attention and linear terms)**.** *Let $\theta$ and $\theta'$ be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
**(i)** *For almost every $h \in \mathbb{R}^{T\times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
**(ii)** *Some $h$ whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm.*
**(iii)** *With an embedding $E$ and a readout $D$ with $ED \neq 0$, the residual network $X \mapsto \big(XE + B_\theta(XE)\big)D$ differs from every skipless network $X \mapsto B_{\theta'}(XE')D'$.*

**Overview.** Two kinds of input suffice. If all tokens are equal, attention returns them unchanged, so the value–output products of all heads must add up to $M$. If one token is zero, all its scores are zero, so every head gives it the average of the tokens, while the skip gives it zero; this asks the same sum to vanish. Part (ii) makes the conflict quantitative, and analyticity turns it into part (i). With an embedding and a readout, the same two inputs give part (iii).

*Proof of Theorem A.12.* Let $F(h) = hM + B_\theta(h) - B_{\theta'}(h)$, and let $\theta'$ have $H'$ heads. The difference $B_{\theta'} - B_\theta$ is one attention block with $H + H'$ heads,

$$B_{\theta'}(h) - B_\theta(h) = \sum_{k=1}^{H+H'} R_k(h)\, h\, C_k,$$

where the $R_k(h)$ are the attention maps of the heads of $\theta'$ and of $\theta$, and the $C_k$ are their value–output products, with a minus sign for the heads of $\theta$. Let $C = \sum_k C_k$.

*Step 1: equal tokens.* Let every row of $h$ be $x$. Each row of $R_k(h)$ sums to 1, so every row of $R_k(h)h$ is $x$. Hence every row of $F(h)$ is $x(M - C)$.

*Step 2: one zero token.* Let the last row of $h$ be 0 and the other rows $y$. Every score in the last row of $R_k(h)$ is a product with the last row of $h$, which is 0. So that row of $R_k(h)$ is $(1/T, \dots, 1/T)$, also with a causal mask, since the last token sees all $T$ tokens. Hence the last row of $F(h)$ is $-\frac{T-1}{T}\, yC$.

*Proof of (ii).* Over $\|x\|, \|y\| \le 1$, Step 1 gives rows of norm up to $\|M - C\|$, and Step 2 up to $\frac{T-1}{T}\|C\| \ge \frac{1}{2}\|C\|$. If $\|M - C\| < \|M\|/3$, then $\|C\| > \frac{2}{3}\|M\|$, so Step 2 reaches more than $\|M\|/3$.

*Proof of (i).* $F$ is real-analytic, because the softmax is, and by (ii) some entry of $F$ is not identically zero. The zeros of a real-analytic function that is not identically zero form a set of measure zero: in one variable they are isolated, and the general case follows by induction on the dimension with Fubini's theorem.

*Proof of (iii).* Let $C_\theta$ and $C_{\theta'}$ be the sums of the value–output products of the heads of $\theta$ and of $\theta'$. On inputs whose rows all equal a row vector $u$, Step 1 gives the rows $uE(I + C_\theta)D$ in the residual network and $uE'C_{\theta'}D'$ in the network without a skip. On inputs whose last row is 0 and whose other rows equal $v$, Step 2 gives the last rows $\frac{T-1}{T}\, vEC_\theta D$ and $\frac{T-1}{T}\, vE'C_{\theta'}D'$. If the two networks are equal, then $E(I + C_\theta)D = E'C_{\theta'}D' = EC_\theta D$, so $ED = 0$. □

We recall Theorem 3.4 from Section 3.

**Theorem 3.4** (Attention blocks)**.** *Let $\theta$ and $\theta'$ be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix. For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*

*Proof of Theorem 3.4.* It is Theorem A.12(i). □

Proposition A.14 follows from a lemma on attention heads.

**Lemma A.13** (Attention heads are identifiable)**.** *Let $T \ge 3$. Let $\Gamma_1, \dots, \Gamma_m \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be distinct, let $C_1, \dots, C_m, M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and write $P_\Gamma(h) = \operatorname{softmax}(h\Gamma h^\top)$, with the softmax taken row by row. If*

$$hM + \sum_{k=1}^{m} P_{\Gamma_k}(h)\, h\, C_k = 0$$

*for all $h$ in a nonempty open subset of $\mathbb{R}^{T \times d_{\text{model}}}$, then $M = 0$ and $C_1 = \cdots = C_m = 0$.*

**Overview.** Move one token along a complex line. Each head becomes a meromorphic function of one complex variable, whose poles lie on a vertical line that its score matrix determines, so heads with distinct score matrices have disjoint poles. At a pole of one head the other terms are regular, so its residue must vanish, and the residues force its value–output product to vanish. What is left is the skip and a uniform head, which two simple inputs separate.

*Proof of Lemma A.13. One moving token.* Fix $h$ in the open set, with rows $x_1, \dots, x_T$, and fix $v \in \mathbb{R}^{1\times d_{\text{model}}}$. Replace the second row by $x_2 + tv$ with $t \in \mathbb{C}$, and look at the first row. For a head with score matrix $\Gamma$, the first row has scores $z_r = x_1 \Gamma x_r^\top$ for $r \neq 2$, and $a + bt$ for $r = 2$, where $a = x_1 \Gamma x_2^\top$ and $b = x_1 \Gamma v^\top$. Its output is

$$g_\Gamma(t) = \frac{e^{a+bt}\,(x_2 + tv) + Z\bar{x}}{e^{a+bt} + Z}, \qquad Z = \sum_{r\neq 2} e^{z_r}, \qquad \bar{x} = \frac{1}{Z}\sum_{r\neq 2} e^{z_r} x_r,$$

where $\bar{x}$ is the average of the other tokens under the weights of the head. The first row of $hM$ is $x_1 M$, which does not depend on $t$.

*Poles.* If $b \neq 0$, then $g_\Gamma$ is meromorphic on $\mathbb{C}$, with simple poles at

$$t_n = c + \frac{(2n+1)\pi i}{b}, \quad n \in \mathbb{Z}, \qquad \text{where } c = \frac{\log Z - a}{b},$$

so all its poles lie on the vertical line $\operatorname{Re} t = c$. The residue at $t_n$ is $(x_2 + t_n v - \bar{x})/b$. If $\Gamma = 0$, then $g_\Gamma$ is affine in $t$ and has no poles.

*The poles tell the heads apart.* For $h$ and $v$ outside a closed set with empty interior, $b \neq 0$ for every $\Gamma_k \neq 0$, and distinct $\Gamma_k \neq 0$ give distinct lines $\operatorname{Re} t = c$. The first condition fails only on the zero sets of nonzero polynomials. For the second, suppose that two heads $\Gamma \neq \Gamma'$ have $c = c'$ for all $h$ and $v$ near some point. Clearing denominators and comparing the terms bilinear in $(x_2, v)$ gives $x_1\Gamma' = \lambda\, x_1\Gamma$ for some number $\lambda \neq 0$. Comparing the terms linear in $v$ then gives $\log Z' = \lambda \log Z$, that is,

$$\log \sum_{r\neq 2} e^{\lambda z_r} \;=\; \lambda \log \sum_{r\neq 2} e^{z_r}.$$

Because $T \geq 3$, the sum has at least two terms, $r = 1$ and $r = 3$, and $z_3$ varies with $x_3$. Differentiating in $z_3$ gives $\operatorname{softmax}(\lambda z)_3 = \operatorname{softmax}(z)_3$ on an interval of $z_3$, which forces $\lambda = 1$. So $x_1\Gamma' = x_1\Gamma$ for all $x_1$ near a point, and $\Gamma' = \Gamma$, a contradiction. With two tokens the sum has the single term $r = 1$, and this step fails (Section 3).

*Residues.* The identity holds for real $t$ near 0, so by analytic continuation it holds on all of $\mathbb{C}$ away from the poles. At a pole $t_n$ of a head $\Gamma_k \neq 0$, no other term is singular, so the residue times $C_k$ vanishes: $(x_2 + t_n v - \bar{x})\, C_k = 0$ for every $n \in \mathbb{Z}$. Subtracting two consecutive ones gives $vC_k = 0$. This holds for all $v$ in an open set, so $C_k = 0$.

*The skip.* What remains is $hM + P_0(h)\, h\, C_0 = 0$, where $C_0$ is the coefficient of $\Gamma = 0$ (or 0 if no $\Gamma_k$ vanishes). This is linear in $h$, so it holds for all $h$. Take the first row of $h$ equal to $x$ and the other rows 0: the first two rows give $x(M + C_0/T) = 0$ and $xC_0/T = 0$. So $C_0 = 0$ and $M = 0$. □

The lemma gives the proposition that Section 3.3 cites.

**Proposition A.14** (The softmaxes must match)**.** *Let $T \geq 3$, let $\theta$ and $\theta'$ be parameters of attention blocks, and suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all $h$ in a nonempty open set. Then $M = 0$, and for every matrix $\Gamma$, the heads with score matrix $\Gamma$ have the same sum of value–output products in $B_\theta$ as in $B_{\theta'}$.*

*Proof of Proposition A.14.* Group the heads of $B_{\theta'}$ and $B_\theta$ by score matrix, add their value–output products, with a minus sign for the heads of $\theta$, and apply Lemma A.13. □

With two tokens, heads can cancel: the softmax of two scores is the logistic function of their difference, and $1/(1+e^{-z}) + 1/(1+e^{z}) = 1$ makes two heads with opposite score matrices add up to a uniform head. A skip still cannot be absorbed when $T = 2$ (Theorem 3.4).

## A.3 Networks under RMSNorm (Theorem 4.4)

We recall Theorem 4.4 from Section 4.

**Theorem 4.4** (Networks under RMSNorm)**.** *Consider pre-RMSNorm networks, with $\varepsilon \geq 0$ and blocks of any kind and width. For almost every choice of weights, a residual network equals no skipless network, whatever its depth, widths and kinds of blocks.*

Write $N_0$ for RMSNorm with $\varepsilon = 0$, which maps each nonzero row $w$ to $\sqrt{d_{\text{model}}}\, w/\|w\|$, with the derivative $\mathrm{D}N_0(w)[b] = \sqrt{d_{\text{model}}}\,\big(b - (b \cdot \hat{w})\,\hat{w}\big)/\|w\|$, where $\hat{w} = w/\|w\|$; it vanishes exactly when $b$ is a multiple of $w$. Outside a set of weights of measure zero, $E$ and $D$ have rank $d_{\text{model}}$, so that $XE$ takes every value in $\mathbb{R}^{T \times d_{\text{model}}}$ and $vD = 0$ only for $v = 0$. Each block is linear in its write, which for an attention block holds the matrices $W_{O,i}$: block $\ell$ adds $a_\ell W^\ell_{\text{write}}$ to the stream, where $a_\ell$, a function of the normalized state $n$ that block $\ell$ reads, holds the activations of its hidden units, or for an attention block the matrices $P_{\Gamma_i}(n)\, n\, W_{V,i}$ of its heads side by side. It depends on the read of block $\ell$ only.

*Proof of Theorem 4.4.* Fix an input $X$ for which $XE$ has no zero row, and let $t \to \infty$.

*The skipless network.* Its hidden states depend on the input only through $N(tXE')$, at its embedding $E'$. On a row $w$ of $XE'$, $N(tw) = w/\sqrt{\|w\|^2/d_{\text{model}} + \varepsilon/t^2}$. For $\varepsilon = 0$ this is $N(w)$, so the output at $tX$ is the output at $X$. For $\varepsilon > 0$ it is a smooth function of $\varepsilon/t^2$ that equals $N_0(w)$ at 0, or 0 if $w = 0$, and the rest of the network is locally Lipschitz, so the output at $tX$ is $G(X) + O(1/t^2)$ for some $G(X)$.

*The residual network.* Let $S = N_0(XE)$ and $\beta(X) = \sum_\ell a_\ell(S) W^\ell_{\text{write}}$, the sum of the outputs of all blocks at the normalized state $S$. By induction over the blocks, $h_\ell = tXE + \sum_{k \leq \ell} a_k(S) W^k_{\text{write}} + O(1/t)$: each hidden state is $tXE + O(1)$, so for large $t$ it has no zero row, the next block reads $N(h_\ell) = S + O(1/t)$, and, being locally Lipschitz, it adds its output at $S$ up to $O(1/t)$. At the readout, on each row, $N(tw + b + O(1/t)) = N_0\big(w + b/t + O(1/t^2)\big) + O(1/t^2) = N_0(w) + \mathrm{D}N_0(w)[b]/t + O(1/t^2)$. So the output at $tX$ is

$$N_0(XE)D + \rho(X)/t + O(1/t^2), \qquad \rho(X) = \mathrm{D}N_0(XE)\big[\beta(X)\big]\, D,$$

with $\mathrm{D}N_0$ taken row by row.

*Comparison.* Let the two networks be equal. Both are continuous on the open set of inputs at which both are defined, so they agree on all of it. The skipless network is defined at $tX$ if and only if it is defined at $X$, by the case $\varepsilon = 0$ above, and everywhere if $\varepsilon > 0$; the residual network is defined at $tX$ for all large $t$. So if the skipless network is defined at $X$, their outputs at $tX$ agree for all large $t$, and $\rho(X) = 0$. These inputs $X$ are dense, and $\rho$ is continuous on the inputs for which $XE$ has no zero row, so $\rho = 0$ there. As $D$ has rank $d_{\text{model}}$, each row of $\beta(X)$ is then a multiple of the same row of $XE$. Here $\beta(X)$ depends on $X$ only through $S$, and as $XE$ takes every value without zero rows, $S$ takes every value whose rows have norm $\sqrt{d_{\text{model}}}$. So equality requires that for every such $S$, each row of $\sum_\ell a_\ell(S) W^\ell_{\text{write}}$ is a multiple of the same row of $S$.

*Measure zero.* Fix all weights but the writes. The condition is linear in the writes, so the writes that satisfy it form a linear subspace. The subspace is proper if some $a_\ell(S)$ has a nonzero entry $(r, j)$ at some $S$: the write of block $\ell$ whose row $j$ is not a multiple of row $r$ of $S$, with all other entries and all other writes zero, violates the condition. And $a_1$ vanishes for all $S$ only if the read of block 1 has a zero column, which happens on a set of measure zero. The weights that satisfy the condition form a closed set, since every $a_\ell$ is continuous in the read and in $S$, and by Fubini's theorem this set has measure zero. □

## A.4 Skipless networks as limits of residual networks (Proposition 5.1)

**Proposition 5.1** (Skipless networks as limits of residual networks)**.** *Let $F$ be a skipless network whose blocks are locally Lipschitz.*

**(i) Without normalization.** *The network $F$ is a limit of residual networks with blocks of the same kinds and widths, uniformly on bounded sets of inputs.*

**(ii) Under pre-RMSNorm with $\varepsilon = 0$.** *The network $F$ is a limit of residual networks with blocks of the same kinds and widths, uniformly on compact sets of inputs at which $F$ is defined.*

Let $F$ have the embedding $E$, the blocks $B_\ell$, with reads $W^\ell_{\text{read}}$ and writes $W^\ell_{\text{write}}$, and the readout $D$, and let $s_0 = XE$ and $s_\ell = B_\ell(s_{\ell-1})$ be its hidden states, so that $F(X) = s_L D$. For $t \geq 1$, let $F_t$ be the residual network with the embedding $E$, the reads $t^{1-\ell} W^\ell_{\text{read}}$, the writes $t^\ell W^\ell_{\text{write}}$ and the readout $t^{-L} D$; under pre-RMSNorm with $\varepsilon = 0$, let $F_t$ keep the embedding, the reads and the readout of $F$ and have the writes $t^\ell W^\ell_{\text{write}}$. Let $r_0 = 0$ and

$$r_\ell \;=\; B_\ell(s_{\ell-1} + r_{\ell-1}) - B_\ell(s_{\ell-1}) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}). \tag{9}$$

The proofs show that the hidden states of $F_t$ are $t^\ell(s_\ell + r_\ell)$ and bound $r_\ell$, which shows that $F_t$ tends to $F$. Norms are Frobenius norms for hidden states and operator norms for matrices.

*Proof of (i).* Block $\ell$ of $F_t$ has its reads scaled by $t^{1-\ell}$ and its writes by $t^\ell$, so it maps $h$ to $t^\ell B_\ell(t^{1-\ell} h)$. We show by induction that its hidden states are $h_\ell = t^\ell(s_\ell + r_\ell)$, with $r_0 = 0$ and $r_\ell$ given by (9). For $\ell = 0$, $h_0 = XE = s_0$. If $h_{\ell-1} = t^{\ell-1}(s_{\ell-1} + r_{\ell-1})$, then

$$h_\ell \;=\; h_{\ell-1} + t^\ell B_\ell(t^{1-\ell} h_{\ell-1}) \;=\; t^\ell\big(B_\ell(s_{\ell-1} + r_{\ell-1}) + t^{-1}(s_{\ell-1} + r_{\ell-1})\big) \;=\; t^\ell(s_\ell + r_\ell).$$

Let $\mathcal{X}$ be a bounded set of inputs and $\sigma_\ell = \sup_{X \in \mathcal{X}} \|s_\ell\|$, which is finite because the blocks are continuous. Suppose that $\|r_{\ell-1}\| \leq C_{\ell-1}/t$ for all $X \in \mathcal{X}$ and $t \geq 1$, which holds for $\ell = 1$ with $C_0 = 0$. Then $s_{\ell-1}$ and $s_{\ell-1} + r_{\ell-1}$ lie in the ball of radius $\sigma_{\ell-1} + C_{\ell-1}$, on which $B_\ell$ is Lipschitz with some constant $\Lambda_\ell$, since $B_\ell$ is locally Lipschitz. Hence

$$\|r_\ell\| \;\leq\; \Lambda_\ell \|r_{\ell-1}\| + t^{-1}\big(\sigma_{\ell-1} + \|r_{\ell-1}\|\big) \;\leq\; C_\ell/t, \qquad C_\ell = \Lambda_\ell C_{\ell-1} + \sigma_{\ell-1} + C_{\ell-1}.$$

Finally $F_t(X) = t^{-L} h_L D = F(X) + r_L D$, so $\|F_t(X) - F(X)\| \leq C_L \|D\|/t$ for all $X \in \mathcal{X}$ and $t \geq 1$. □

*Proof of (ii).* Let $\mathcal{X}$ be a compact set of inputs at which $F$ is defined, that is, at which no row of $XE$ or of a hidden state of $F$ vanishes. Here $N$ divides each row of $h$ by its norm and multiplies it by $\sqrt{d_{\text{model}}}$, so $N(ch) = N(h)$ for $c > 0$, and $\|N(y') - N(y)\| \leq 2\sqrt{d_{\text{model}}}\,\|y' - y\|/\|y\|$ for nonzero rows $y, y'$. The hidden states of $F$ are $s_0 = XE$ and $s_\ell = B_\ell(N(s_{\ell-1}))$, and $F(X) = N(s_L)D$. As no row of $s_\ell$ vanishes on $\mathcal{X}$, continuity and compactness give $\delta > 0$ below the norm of every row of every $s_\ell$, $\ell \geq 1$, on $\mathcal{X}$. Since $N$ removes the factor $t^{\ell-1}$, the hidden states of $F_t$ are again $h_0 = s_0$ and $h_\ell = t^\ell(s_\ell + r_\ell)$, now with $r_0 = 0$ and

$$r_\ell \;=\; B_\ell\big(N(s_{\ell-1} + r_{\ell-1})\big) - B_\ell\big(N(s_{\ell-1})\big) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}),$$

as long as no row of $s_{\ell-1} + r_{\ell-1}$ vanishes. The maps $B_\ell$ are Lipschitz on the bounded image of $N$, with some constants $\Lambda_\ell$, and $\sigma_\ell = \sup_{X \in \mathcal{X}} \|s_\ell\|$ is finite. If every row of $r_{\ell-1}$ is shorter than $\delta$, then $\|r_\ell\| \leq 2\sqrt{d_{\text{model}}}\,\Lambda_\ell \|r_{\ell-1}\|/\delta + t^{-1}(\sigma_{\ell-1} + \|r_{\ell-1}\|)$, so by induction $\|r_\ell\| \leq C_\ell/t$ on $\mathcal{X}$ for $t \geq 1$, with $C_0 = 0$ and $C_\ell = (2\sqrt{d_{\text{model}}}\,\Lambda_\ell/\delta + 1)C_{\ell-1} + \sigma_{\ell-1}$, and every row of $r_\ell$ is shorter than $\delta$ once $t \geq t_0 = \max(1, \max_\ell 2C_\ell/\delta)$. Finally $\|F_t(X) - F(X)\| = \|(N(s_L + r_L) - N(s_L))D\| \leq 2\sqrt{d_{\text{model}}}\,\|D\|\,C_L/(\delta t)$ for all $X \in \mathcal{X}$ and $t \geq t_0$. □

*Remark* A.15 (The rate). The proofs give the rate $1/t$. Without normalization, for every bounded set $\mathcal{X}$ of inputs there is $C$ such that $\|F_t(X) - F(X)\| \leq C/t$ for all $X \in \mathcal{X}$ and $t \geq 1$. Under pre-RMSNorm with $\varepsilon = 0$, the same holds for all $t \geq t_0$, for every compact set $\mathcal{X}$ of inputs at which $F$ is defined, with $t_0$ depending on $\mathcal{X}$. The constants depend on $\mathcal{X}$ and on $F$.

# B Universal approximation: width and biases

Lin and Jegelka (2018) show that residual ReLU networks whose blocks have a single hidden unit, $h \mapsto h + \mathrm{ReLU}(hw + c)\,v$ with $w \in \mathbb{R}^{d\times 1}$, $v \in \mathbb{R}^{1\times d}$ and a bias $c \in \mathbb{R}$, approximate every integrable function on $\mathbb{R}^d$ in $L^1$ as the depth grows, while ReLU networks without skips whose hidden layers have width at most the input dimension $d$ cannot (their Proposition 2.1). The separation is confined to this narrow regime: with hidden layers of width $d + 1$, ReLU networks without skips already approximate every continuous function uniformly on compact sets (Hanin and Sellke, 2017).

The construction also needs the biases, for the shifts and the cut-offs at constants of its blocks (their Proposition 3.2). Without biases, a ReLU network, with or without skips, is positively homogeneous of degree one, $F(\lambda x) = \lambda F(x)$ for $\lambda > 0$, since ReLU and the linear maps are. On every ray $\{t\theta : t > 0\}$ such a network is a multiple of $t$, and so is every limit of such networks, uniformly or in $L^1$ on a compact set, on almost every ray: the multiples of $t$ form a closed space, and by Fubini $L^1$ convergence passes to almost every ray. In one dimension, for instance, every such network is $x \mapsto a\,\mathrm{ReLU}(x) - b\,\mathrm{ReLU}(-x)$, which stays at $L^1$ distance at least $(2-\sqrt{2})/3$ from $x^2$ on $[-1,1]$. On all of $\mathbb{R}^d$, no integrable function other than 0 can be approximated in $L^1$, since a positively homogeneous function other than 0 is not integrable.

# C Blocks with biases

The main text treats blocks without biases, as in many current language models. With biases the answers change: every ReLU, GELU, ReLU$^2$ and gated block absorbs a linear term with more units, while $\tanh$ and attention blocks still never do (Theorem C.1). To absorb its skip, almost every ReLU, GELU and gated block needs exactly $d_{\text{model}}$ more units, and almost every ReLU$^2$ block between $2d_{\text{model}}$ and $3d_{\text{model}}$ (Theorem C.2). Table 2 collects these results.

**Blocks with biases.** A vanilla MLP block with biases computes $B_\theta(x) = f(xW_{\text{read}} + b)\,W_{\text{write}} + c$, with $\theta = (W_{\text{read}}, b, W_{\text{write}}, c)$, a unit bias $b \in \mathbb{R}^{1\times m}$ and an output bias $c \in \mathbb{R}^{1\times d_{\text{model}}}$; unit $i$ has the read $w_i$, the bias $b_i$ and the write $\hat{w}_i$. In a gated block with biases, unit $j$ has a gate read $g_j$ (a column of $W_{\text{g}}$) and a gate bias $b^{\text{g}}_j$, a value read $v_j$ (a column of $W_{\text{v}}$) and a value bias $b^{\text{v}}_j$, and a write $\hat{w}_j$, and

$$B_\theta(x) = \sum_{j=1}^{m} \sigma\left(xg_j + b^{\text{g}}_j\right)\left(xv_j + b^{\text{v}}_j\right)\hat{w}_j + c.$$

An attention block with biases adds a bias row to every query, key and value and to its output: with $\mathbf{1}$ the column of $T$ ones,

$$B_\theta(h) = \sum_{i=1}^{H} P_i(h)\left(hW_{V,i} + \mathbf{1}b_{V,i}\right)W_{O,i} + \mathbf{1}b_O,$$

where $P_i(h)$ is the softmax, row by row, of $(hW_{Q,i}+\mathbf{1}b_{Q,i})(hW_{K,i}+\mathbf{1}b_{K,i})^\top/\sqrt{d_h}$, with or without a causal mask. These scores do not depend on the positions of the tokens, which excludes rotary embeddings. A block absorbs $xM$, for $M \neq 0$, if a block of the same kind with biases satisfies (2), or for attention, with the same mask, $hM + B_\theta(h) = B_{\theta'}(h)$ for all $h \in \mathbb{R}^{T\times d_{\text{model}}}$.

**Theorem C.1** (Exact absorption with biases)**.** *Let blocks have biases, and let $M \neq 0$.*
**(i)** *Every ReLU, GELU, ReLU$^2$, ReGLU, SwiGLU or GeGLU block absorbs $xM$ with more hidden units.*
**(ii)** *No $\tanh$ block and no attention block absorbs $xM$, at any width.*
**(iii)** *For generic $\theta$, no block $B_{\theta'}$ of the same width as $B_\theta$ satisfies (2).*

Part (iii) follows from (ii) for $\tanh$ and attention blocks, and from the proofs of Theorem C.2 for the others.

**Theorem C.2** (Extra units with biases)**.** *Let blocks have biases, let $M$ be invertible and let $\theta$ be generic.*
**(i)** *A ReGLU, SwiGLU or GeGLU block absorbs $xM$ with $d_{\text{model}}$ more hidden units, and not with fewer.*
**(ii)** *A ReLU or GELU block with at least $d_{\text{model}}$ hidden units absorbs $xM$ with $d_{\text{model}}$ more hidden units, and not with fewer.*
**(iii)** *A ReLU$^2$ block with at least $d_{\text{model}}$ hidden units absorbs $xM$ with $3d_{\text{model}}$ more hidden units, and not with fewer than $2d_{\text{model}}$.*

| **Proven** | **ReLU** | **GELU** | **tanh** | **ReLU$^2$** | **ReGLU, SwiGLU, GeGLU** | **Attention** |
|---|---|---|---|---|---|---|
| one block, same width | measure zero Thm. C.1(iii) | | never Thm. C.1(ii) | measure zero Thm. C.1(iii) | | never Thm. C.1(ii) |
| one block, more hidden units or heads | exactly $d_{\text{model}}$ more Thm. C.2(ii) | | no Thm. C.1(ii) | $2d_{\text{model}}$ to $3d_{\text{model}}$ more Thm. C.2(iii) | exactly $d_{\text{model}}$ more Thm. C.2(i) | no Thm. C.1(ii) |

**Table 2:** What is proven for one block with biases. Never or no: for no weights. Measure zero: only for a null set of weights. Exactly $d_{\text{model}}$ more: for almost every block, $d_{\text{model}}$ more hidden units suffice and fewer do not. Scope as in Section 2.

*Remark* C.3 (Constructions for every block). Every gated block absorbs $xM$ with $\operatorname{rank} M$ more units, and for generic $\theta$ fewer do not (proof of Theorem C.2(i)). Every ReLU or GELU block absorbs $xM$ with $2d_{\text{model}}$ more units, and with $d_{\text{model}}$ when its read has rank $d_{\text{model}}$. Every ReLU$^2$ block absorbs $xM$ with $4d_{\text{model}}$ more units, and with $3d_{\text{model}}$ when its read has rank $d_{\text{model}}$. Almost every ReLU$^2$ block with at least $d_{\text{model}}(d_{\text{model}}+3)/2$ units absorbs it with $d_{\text{model}}(d_{\text{model}}+3)/2$ more, which is fewer than $3d_{\text{model}}$ only for $d_{\text{model}} = 2$, where it is 5.

*Remark* C.4 (Bounded sets of inputs). GELU, SwiGLU, GeGLU, $\tanh$ and attention blocks are real-analytic in their input, so an equality on a nonempty open set of inputs holds everywhere, and Theorems C.1 and C.2 hold on every such set. ReLU, ReLU$^2$ and ReGLU blocks are not. On a ball $\mathcal{X}$ of inputs, a unit whose argument (for ReGLU, whose gate's argument) does not vanish on $\mathcal{X}$ is a polynomial there. If $d_{\text{model}}$ units of a ReLU block with linearly independent reads, or $d_{\text{model}}(d_{\text{model}}+3)/2$ generic units of a ReLU$^2$ or ReGLU block, are active on all of $\mathcal{X}$, changing their writes and the output bias absorbs $xM$ on $\mathcal{X}$ at the same width. So on $\mathcal{X}$, the entries "measure zero" of Table 2 fail for ReLU, ReLU$^2$ and ReGLU blocks, for an open set of blocks. The lower bounds below use inputs of unbounded size: rays $x = tu$ with $t \to \infty$, or the hyperplanes on which the arguments of the units vanish, wherever they lie.

*Remark* C.5 (Networks). Without normalization, absorbing the skip block by block, with $M = I$, shows that every residual network of ReLU$^2$, ReGLU, SwiGLU or GeGLU blocks with biases equals a skipless network with $4d_{\text{model}}$ (ReLU$^2$) or $d_{\text{model}}$ (gated) more units per block; without biases, none with $ED \neq 0$ does (Theorem 4.1(i)). As without biases, every residual ReLU or GELU network whose reads have rank $d_{\text{model}}$ equals a skipless one with $d_{\text{model}}$ more units per block. A skipless network of $\tanh$ blocks with biases is bounded, so it equals no residual one with $ED \neq 0$. Networks with biases under pre-RMSNorm are not treated here.

**Open questions.** The exact number of extra units for ReLU$^2$ blocks lies between $2d_{\text{model}}$ and $3d_{\text{model}}$; for $d_{\text{model}} = 2$, five suffice once the block has five units (Remark C.3). Attention with biases and rotary embeddings is open.

## C.1 Constructions and blocks that never absorb

Let $e_a$ be the $a$-th standard basis column of $\mathbb{R}^{d_{\text{model}}}$ and $M_a$ the $a$-th row of $M$; then $xM$ is the sum over $a$ of $(xe_a)\, M_a$.

*Proof of Theorem C.1(i).* The constructions also give the upper bounds of Theorem C.2. Each construction keeps the units of $B_\theta$, changes some of their writes and the output bias, and adds units.

*Gated.* A unit with gate read 0 and gate bias 1 computes $\sigma(1)\,(xv + b^{\text{v}})\,\hat{w}$, which is affine in $x$, and $\sigma(1) \neq 0$ for ReLU, SiLU and GELU. Write $M$ as a sum of $\operatorname{rank} M$ matrices of rank one, each a column times a row. For each term, a unit with gate read 0, gate bias 1, value bias 0, the column as its value read and the row divided by $\sigma(1)$ as its write adds $x$ times the term. So $\operatorname{rank} M \leq d_{\text{model}}$ units add $xM$.

*ReLU and GELU.* Both satisfy (3). Two new units with reads $\pm e_a$, biases 0 and writes $\pm M_a$ add $(xe_a)\, M_a$, so $2d_{\text{model}}$ new units add $xM$. If the read has rank $d_{\text{model}}$, choose $d_{\text{model}}$ units with linearly independent reads, and let the rows $\gamma_i$, for these units $i$, solve $\sum_i w_i \gamma_i = M$. Give unit

$i$ the write $\hat{w}_i + \gamma_i$, and add a unit with read $-w_i$, bias $-b_i$ and write $-\gamma_i$. By (3) the two add $(xw_i + b_i)\,\gamma_i$, and subtracting $\sum_i b_i\gamma_i$ from the output bias leaves $xM$, with $d_{\text{model}}$ new units.

*ReLU*$^2$. Let $r(z) = \max(z, 0)^2$, so that $r(z) + r(-z) = z^2$. Four new units compute $r\big(\pm(xe_a + 1)\big)$, with the write $M_a/4$, and $r\big(\pm(xe_a - 1)\big)$, with the write $-M_a/4$. Together they add $\big((xe_a + 1)^2 - (xe_a - 1)^2\big)M_a/4 = (xe_a)\,M_a$, so $4d_{\text{model}}$ new units add $xM$. If the read has rank $d_{\text{model}}$, take the units $i$ and the rows $\gamma_i$ as above, and let $z_i = xw_i + b_i$. Give unit $i$ the write $\hat{w}_i + \gamma_i$ and add three units: one with read $-w_i$, bias $-b_i$ and write $\gamma_i$, and two with reads $\pm w_i$, biases $\pm(b_i - \frac{1}{2})$ and writes $-\gamma_i$. Together they add $\gamma_i\big(z_i^2 - (z_i - \frac{1}{2})^2\big) = (xw_i)\,\gamma_i + (b_i - \frac{1}{4})\,\gamma_i$, and the output bias absorbs the constants: $3d_{\text{model}}$ new units. Finally, for $d_{\text{model}}(d_{\text{model}} + 3)/2$ units $i$ of the block and new rows $\gamma_i$, give unit $i$ the write $\hat{w}_i + \gamma_i$ and add a unit with read $-w_i$, bias $-b_i$ and write $\gamma_i$: these add $\sum_i \gamma_i z_i^2$. For each output coordinate $o$, $\sum_i \gamma_i z_i^2 = xM + \text{const}$ is a linear system of $d_{\text{model}}(d_{\text{model}} + 3)/2$ equations, one for each coefficient of a polynomial of degree two other than the constant, in the $d_{\text{model}}(d_{\text{model}} + 3)/2$ unknowns $\gamma_{i,o}$. For generic $\theta$ it has a solution. □

*Proof of Theorem C.1(ii).* tanh. $B_{\theta'} - B_\theta$ is bounded, and $xM$ is not.

*Attention.* Suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all $h$, and let $C$ and $C'$ be the sums of the value–output products $W_{V,i}W_{O,i}$ of the heads of $\theta$ and of $\theta'$. If all rows of $h$ equal $x$, all keys of a head are equal, so each row of $P_i(h)$ is uniform on the tokens it sees, and all values equal $xW_{V,i} + b_{V,i}$. So every row of $B_\theta(h)$ is $xC$ plus a constant row, and the equation gives $M = C' - C$. Now add $\epsilon\delta$ to the first row of this $h$, for a row $\delta$, and look at the second row, which sees the first token with or without a causal mask. The second row of $hM$ does not change. In the second row of $B_\theta(h)$, the derivatives of the attention weights at $\epsilon = 0$ sum to zero over the tokens and multiply equal values, so the derivative in $\epsilon$ at $\epsilon = 0$ is $\delta C/T_2$, with $T_2$ the number of tokens that the second token sees. Hence $\delta C = \delta C'$ for every $\delta$, so $C = C'$ and $M = 0$, a contradiction. □

## C.2 Lower bounds

For a unit $j$ of a gated block with biases, let $\Delta_j = \big(b_j^{\text{g}} v_j + b_j^{\text{v}} g_j\big)\,\hat{w}_j$, a $d_{\text{model}} \times d_{\text{model}}$ matrix of rank at most one.

**Lemma C.6** (Gated blocks with biases)**.** *Let ReGLU, SwiGLU or GeGLU blocks $B_\theta$ and $B_{\theta'}$ with biases satisfy (2), where $B_{\theta'}$ has $n$ more units than $B_\theta$. Let the gate reads of $B_\theta$ be nonzero and pairwise non-collinear, and its value reads and writes nonzero. Then for some set $S$ of units of $B_\theta$, the matrix $M + \sum_{i\in S} \Delta_i$ is a sum of at most $n$ matrices of rank one.*

*Proof. Outline.* Along rays $x = tu$ with $t \to \infty$, every gate becomes a ReLU gate. The terms of order $t^2$ and $t$ of (2) then split hyperplane by hyperplane, and those of order $t$ express $M$ through the units on one side of each hyperplane.

*Expansion along rays.* Group the units of both blocks with a nonzero gate read $g$ by the hyperplane $\mathcal{H} = \{x : xg = 0\}$, choose a unit normal $\nu$ of $\mathcal{H}$, and write $g = s\eta\nu$ with $\eta > 0$ and a side $s = \pm 1$. Let $[\,\cdot\,]$ be 1 if the condition inside holds and 0 otherwise. For $x \notin \mathcal{H}$ and $t \to \infty$,

$$\sigma\big(txg + b^{\text{g}}\big) = t\eta\,\text{ReLU}(s\,x\nu) + b^{\text{g}}\,[s\,x\nu > 0] + o(1/t),$$

exactly for large $t$ when $\sigma$ is ReLU, and up to terms that decay exponentially in $t$ when $\sigma$ is SiLU or GELU. So the unit contributes

$$t^2\,\text{ReLU}(s\,x\nu)\,(xv)\,\eta\hat{w} + t\,\big(\text{ReLU}(s\,x\nu)\,\eta b^{\text{v}}\hat{w} + [s\,x\nu > 0]\,(xv)\,b^{\text{g}}\hat{w}\big) + O(1),$$

and a unit with gate read 0 contributes $t\,\sigma(b^{\text{g}})\,(xv)\,\hat{w} + O(1)$. Put $tx$ into (2) and compare the coefficients of $t^2$ and of $t$ at each $x$ off all the hyperplanes. For a hyperplane $\mathcal{H}$ and a side $\pm$, let $A^\pm$, $a^\pm$ and $G^\pm$ be the sums of $\eta v\hat{w}$, $\eta b^{\text{v}}\hat{w}$ and $b^{\text{g}}v\hat{w}$ over the units of $B_{\theta'}$ on that side, minus the same sums over the units of $B_\theta$, and let $Z$ be the same difference for the sums of $\sigma(b^{\text{g}})\,v\hat{w}$ over the units with gate read 0. Then, with sums over the hyperplanes,

$$\begin{aligned} 0 &= \textstyle\sum_{\mathcal{H}} \big(\text{ReLU}(x\nu)\,xA^+ + \text{ReLU}(-x\nu)\,xA^-\big), \\ xM &= \textstyle\sum_{\mathcal{H}} \big(\text{ReLU}(x\nu)\,a^+ + \text{ReLU}(-x\nu)\,a^- + [x\nu > 0]\,xG^+ + [x\nu < 0]\,xG^-\big) + xZ. \end{aligned}$$

*One hyperplane at a time.* Near a point of $\mathcal{H}$ that lies on no other hyperplane, the terms of the other hyperplanes are smooth. In the second identity, continuity across $\mathcal{H}$ forces $yG^+ = yG^-$ for every row $y$ with $y\nu = 0$, so $G^+ - G^- = \nu q$ for a row $q$. Then $[x\nu > 0]\, x\nu q = \mathrm{ReLU}(x\nu)\, q$, and since the identity has no kink at $\mathcal{H}$, $a^+ + q + a^- = 0$. The terms of $\mathcal{H}$ are then $x\Delta_{\mathcal{H}}$, with

$$\Delta_{\mathcal{H}} = G^- - \nu a^- = G^+ + \nu a^+,$$

and the second identity becomes $M = \sum_{\mathcal{H}} \Delta_{\mathcal{H}} + Z$. As $g = \pm\eta\nu$ on side $\pm$, $G^{\pm} \pm \nu a^{\pm}$ is the sum of the matrices $\Delta_j$ over the units of $B_{\theta'}$ on side $\pm$, minus the same sum over the units of $B_\theta$. In the first identity, the same argument applied to $\mathrm{ReLU}(\pm x\nu) = (|x\nu| \pm x\nu)/2$ gives $A^+ + A^- = \nu p$ for a row $p$, and then $p = 0$, since $|x\nu|\, x\nu$ has a jump in its second derivative. So $A^+ + A^- = 0$.

*Counting.* Orient the hyperplane $\mathcal{H}_i$ of unit $i$ of $B_\theta$ so that unit $i$ lies on side $+$; no other unit of $B_\theta$ lies on $\mathcal{H}_i$. Let $J^{\pm}$ be the sets of units of $B_{\theta'}$ on side $\pm$ of $\mathcal{H}_i$. If both were empty, $A^+ + A^-$ would be $-\eta_i v_i \hat{w}_i$, where $g_i = \eta_i \nu$, which does not vanish; so $\mathcal{H}_i$ carries $|J^+| + |J^-| - 1 \geq 0$ extra units. If $J^+$ is empty, $\Delta_{\mathcal{H}_i} = -\Delta_i$. Otherwise $\Delta_{\mathcal{H}_i}$ is the sum of the $\Delta_j$ over $J^-$, at most $|J^-| \leq |J^+| + |J^-| - 1$ matrices of rank one. A hyperplane of no unit of $B_\theta$ carries $|J^+| + |J^-|$ units of $B_{\theta'}$, with $J^{\pm}$ defined in the same way, and $\Delta_{\mathcal{H}}$ is a sum of at most that many matrices of rank one. Each unit of $B_{\theta'}$ with gate read 0 adds one matrix of rank one to $Z$, and $B_\theta$ has no such unit. With $S$ the units $i$ for which $J^+$ is empty, $M + \sum_{i \in S} \Delta_i$ is therefore a sum of at most $n$ matrices of rank one. □

*Proof of Theorem C.2(i).* The argument also gives Theorem C.1(iii) for gated blocks. The construction above gives $d_{\mathrm{model}}$ units. For each set $S$ of units of $B_\theta$, the determinant of $M + \sum_{i \in S} \Delta_i$ is a polynomial in $\theta$, and it equals $\det M \neq 0$ when all biases vanish. So for generic $\theta$ it is nonzero for each of the finitely many sets $S$, the hypotheses of Lemma C.6 hold, and the lemma gives $n \geq d_{\mathrm{model}}$. For $M \neq 0$ of any rank, a minor of size $\mathrm{rank}\, M$ that does not vanish at $M$ plays the role of the determinant, and gives $n \geq \mathrm{rank}\, M \geq 1$. □

*Proof of Theorem C.2(ii).* The argument also gives Theorem C.1(iii) for ReLU and GELU blocks. The construction above gives $d_{\mathrm{model}}$ units. For the converse, let $B_{\theta'}$ satisfy (2). ReLU changes by at most $|b|$ when its argument changes by $b$, and GELU differs from ReLU by at most $\kappa < 0.17$ (proof of Theorem 4.2). So for both activations and all $t > 0$,

$$\big|\sigma(tz + b) - t\,\mathrm{ReLU}(z)\big| \leq |b| + \kappa,$$

and $\sigma(tz + b)/t \to \mathrm{ReLU}(z)$ as $t \to \infty$. Put $tx$ into (2), divide by $t$, and let $t \to \infty$: the ReLU blocks without biases that have the reads and writes of $B_\theta$ and $B_{\theta'}$ satisfy (2). For generic $\theta$ these reads and writes are generic, so $B_{\theta'}$ has at least $d_{\mathrm{model}}$ more units by Proposition A.11(ii), and for any $M \neq 0$ it is not of the same width, by Theorem 3.1(i). □

*Proof of Theorem C.2(iii).* The argument also gives Theorem C.1(iii) for $\mathrm{ReLU}^2$ blocks. The construction above gives $3d_{\mathrm{model}}$ units. For the converse, let $B_{\theta'}$, with $n$ more units than $B_\theta$, satisfy (2), and let $r(z) = \max(z, 0)^2$.

*Squares.* Group the units of both blocks with a nonzero read by the hyperplane $\mathcal{H} = \{x : x\omega + \omega_0 = 0\}$ on which their argument vanishes, with a representative $(\omega, \omega_0)$ and a side: a unit with read $\lambda\omega$ and bias $\lambda\omega_0$ computes $\lambda^2\, r\big(\pm(x\omega + \omega_0)\big)$ times its write, on side $\pm$ as $\lambda \gtrless 0$. Let $\mu^{\pm}$ be the sum of $\lambda^2$ times the write over the units of $B_{\theta'}$ on side $\pm$, minus the same sum over the units of $B_\theta$. By (2), $B_{\theta'} - B_\theta$ is a polynomial. Near a point of $\mathcal{H}$ on no other hyperplane, the terms of $\mathcal{H}$ are $\mu^+ r(z) + \mu^- r(-z)$, with $z = x\omega + \omega_0$, whose second derivative in the direction of $\omega$ jumps across $\mathcal{H}$ by a nonzero multiple of $\mu^+ - \mu^-$. So $\mu^+ = \mu^- =: \mu_{\mathcal{H}}$, the terms of $\mathcal{H}$ are $\mu_{\mathcal{H}}\, z^2$, and

$$\textstyle\sum_{\mathcal{H}} \mu_{\mathcal{H}}\, (x\omega_{\mathcal{H}} + \omega_{0,\mathcal{H}})^2 = xM + \mathrm{const}, \tag{10}$$

with the representatives $(\omega_{\mathcal{H}}, \omega_{0,\mathcal{H}})$.

*Costs.* For generic $\theta$ the units of $B_\theta$ have nonzero reads and writes and lie on distinct hyperplanes. Take unit $i$ itself, $(w_i, b_i)$, as the representative of its hyperplane $\mathcal{H}_i$, on side $+$. Then the sums of $\lambda^2$

times the writes of the units of $B_{\theta'}$ on $\mathcal{H}_i$ are $\hat{w}_i + \mu_{\mathcal{H}_i}$ on side $+$ and $\mu_{\mathcal{H}_i}$ on side $-$, so $B_{\theta'}$ has at least one unit on $\mathcal{H}_i$, and at least two unless $\mu_{\mathcal{H}_i} \in \{0, -\hat{w}_i\}$. On a hyperplane of no unit of $B_\theta$, it has at least two units if $\mu_{\mathcal{H}} \neq 0$. Let $S$ and $U$ be the sets of units $i$ of $B_\theta$ with $\mu_{\mathcal{H}_i} = -\hat{w}_i$ and with $\mu_{\mathcal{H}_i} \notin \{0, -\hat{w}_i\}$, and $\mathcal{N}$ the set of the other hyperplanes with $\mu_{\mathcal{H}} \neq 0$. Then $n \geq |U| + 2|\mathcal{N}|$.

*Dimensions.* Modulo constants, a polynomial of degree at most two is a pair of a quadratic form and a linear part, in a space of dimension $d_{\text{model}}(d_{\text{model}} + 3)/2$, in which $(x\omega + \omega_0)^2$ becomes $\varphi(\omega, \omega_0) = (\omega\omega^\top, 2\omega_0\omega)$. For each output coordinate $o$, with $M_{:,o}$ the column $o$ of $M$, (10) reads

$$\textstyle\sum_{i \in U} \mu_{\mathcal{H}_i, o}\, \varphi(w_i, b_i) + \sum_{\mathcal{H} \in \mathcal{N}} \mu_{\mathcal{H}, o}\, \varphi(\omega_{\mathcal{H}}, \omega_{0,\mathcal{H}}) = \tau_o,$$

where $\tau_o = (0, M_{:,o}) + \sum_{i \in S} \hat{w}_{i,o}\, \varphi(w_i, b_i)$. So the vectors $\tau_1, \ldots, \tau_{d_{\text{model}}}$ and $\varphi(w_i, b_i)$, $i \in U$, lie in the span of $|U| + |\mathcal{N}|$ vectors.

*Genericity.* For each of the finitely many pairs of disjoint sets $S$ and $U$ of units of $B_\theta$, these $d_{\text{model}} + |U|$ vectors span a space of dimension at least $\min\big(d_{\text{model}}(d_{\text{model}} + 3)/2, d_{\text{model}} + |U|\big)$ for generic $\theta$. The condition is that some minor of the matrix with these columns, a polynomial in $\theta$, does not vanish. It holds when the writes $\hat{w}_i$, $i \in S$, vanish and the reads are generic: then the $\tau_o = (0, M_{:,o})$ span all linear parts, as $M$ is invertible, and for generic columns $w_i$, any $\min\big(|U|, d_{\text{model}}(d_{\text{model}} + 1)/2\big)$ of the forms $w_i w_i^\top$ are linearly independent.

*Conclusion.* So $|U| + |\mathcal{N}| \geq \min\big(d_{\text{model}}(d_{\text{model}} + 3)/2, d_{\text{model}} + |U|\big)$. If the minimum is $d_{\text{model}} + |U|$, then $|\mathcal{N}| \geq d_{\text{model}}$ and $n \geq 2d_{\text{model}}$. Otherwise $|U| + |\mathcal{N}| \geq d_{\text{model}}(d_{\text{model}} + 3)/2 \geq 2d_{\text{model}}$, and again $n \geq 2d_{\text{model}}$. For $M \neq 0$ of any rank, the same argument with $\operatorname{rank} M$ in place of $d_{\text{model}}$ gives $n \geq 2$. □